

TONGJI CALCULUS, 7TH EDITION · CHAPTER 1

FUNCTIONS & LIMITS

DETAILED SOLUTIONS

100 original and textbook-method adaptations · basic to challenge

Goodnotes-optimized 4:3 edition · 2026



How to use this set

100 Questions	All 10 sections Sections	Basic → challenge Progression	8 mixed formats Formats
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Work through Foundation, Methods, Synthesis, and Challenge in that order. Each question shows its difficulty, format, and suggested time. Attempt it before opening the solution book; record the cause of each error and retry after 48 hours.

“Textbook-method adaptation” means the representative method is retained while the structure, parameters, or task have been independently redesigned.

Scope: Chapter 1 tools only. Derivatives, L'Hopital's rule, Taylor expansions, and power series are not used.

Coverage matrix

1	Mappings and Functions	11	domains, composition, inverses, properties
2	Limits of Sequences	10	epsilon-N, properties, subsequences, recursion
3	Limits of Functions	10	one-sided limits, epsilon-delta, properties
4	Infinitesimals and Infinite Quantities	7	infinitesimals, infinity, reciprocals
5	Algebra of Limits	13	factoring, rationalization, parameters
6	Existence Criteria and Two Fundamental Limits	16	squeeze, monotone-bounded, two limits
7	Comparison of Infinitesimals	11	orders, equivalents, cancellation traps
8	Continuity and Discontinuities	10	continuity, parameters, classification
9	Operations on Continuous Functions	4	operations, composition, elementary functions
10	Continuous Functions on Closed Intervals	8	extrema, zeros, IVT, uniform continuity

Foundation

Attempt independently before consulting the solutions.

§1 · Mappings and Functions

Basic

Single choice

4 min

Q001 · Identify a Valid Mapping

Question

Let $X=\{-1,0,1\}$ and $Y=\{0,1\}$. Which correspondence defines a function from X to Y ?

Answer first

A

Knowledge points

- every domain element must have an image
- each input must have a unique output
- every output must belong to the codomain

Reading the problem and choosing a method

Check the three defining requirements one by one: every element of X must be covered, its image must be unique, and every image must lie in Y . Failure of any one condition invalidates the correspondence.

Step-by-step derivation

Step 1. For A, $(-1)^2=1$, $0^2=0$, and $1^2=1$, so all three inputs have exactly one image.

Step 2. All outputs in A are either 0 or 1, hence they belong to Y ; therefore A is valid.

Step 3. B leaves $f(0)$ undefined, so it does not cover the whole domain X .

Step 4. C gives the same input 1 two outputs, while D gives $f(1)=2 \notin Y$; therefore C and D are invalid.

Common pitfalls

- checking only whether a formula can be evaluated and ignoring the prescribed codomain
- allowing one input to have two different outputs

Verification

Substitution in A gives the ordered-pair set $\{(-1,1),(0,0),(1,1)\}$. Every first component appears exactly once, and every second component lies in Y.

Method summary

A mapping is a function only when it is defined everywhere on the domain, single-valued, and codomain-respecting.

Extension prompt

If the codomain in D were changed to $\{0,1,2\}$, then D would define a function from X to that new codomain.

§1 · Mappings and Functions

Basic

Fill in the blank

5 min

Textbook-method adaptation

Q002 · Common Domain of a Radical and a Logarithm

Question

The natural domain of $f(x)=\sqrt{2-x}+\ln(x+1)$ is _____.

Answer first

$(-1,2]$

Knowledge points

- an even root requires a nonnegative radicand
- the argument of a natural logarithm must be strictly positive
- simultaneous restrictions are intersected

Reading the problem and choosing a method

Because both expressions occur in the same function, x must satisfy both restrictions. The radical may include its zero endpoint, whereas the logarithm may not.

Step-by-step derivation

Step 1. For $\sqrt{2-x}$ to be defined, require $2-x \geq 0$, hence $x \leq 2$.

Step 2. For $\ln(x+1)$ to be defined, require $x+1 > 0$, hence $x > -1$.

Step 3. Intersect $(-1, +\infty)$ with $(-\infty, 2]$.

Step 4. The intersection is $(-1, 2]$; -1 is excluded and 2 is included.

Common pitfalls

- replacing the logarithmic condition $x+1 > 0$ with $x+1 \geq 0$
- listing the two restrictions without taking their intersection

Verification

At $x=2$ the value $\sqrt{0+\ln 3}$ is defined, while $x=-1$ would produce $\ln 0$, confirming the stated endpoint types.

Method summary

For a natural domain, collect every restriction, intersect them, and distinguish strict from non-strict inequalities.

Extension prompt

If $\ln(x+1)$ were replaced by $1/\ln(x+1)$, one would also require $\ln(x+1) \neq 0$, excluding $x=0$.

§1 · Mappings and Functions

Basic

True/false with justification

3 min

Q003 · A Necessary Condition for Composition

Question

True or false: If $(f \circ g)(x_0)$ is defined, then $g(x_0)$ must belong to the domain of f .

Answer first

True.

Knowledge points

- definition of a composite function
- input condition for the outer function

Reading the problem and choosing a method

$(f \circ g)(x_0)$ means $f(g(x_0))$. For this expression to exist, $g(x_0)$ must first exist and must then be an admissible input for f .

Step-by-step derivation

Step 1. By definition, $(f \circ g)(x_0) = f(g(x_0))$.

Step 2. Because the composite value is defined, x_0 must lie in the domain of g , so $g(x_0)$ exists.

Step 3. For $f(g(x_0))$ to be evaluated, the input $g(x_0)$ must lie in the domain of f .

Step 4. Thus the stated condition is necessary, and the statement is true.

Common pitfalls

- requiring x_0 itself to lie in both domains instead of checking the inner output $g(x_0)$
- reversing the order of composition

Verification

If $g(x_0)$ were outside the domain of f , then $f(g(x_0))$ would be undefined, contradicting the hypothesis.

Method summary

For a composition, inspect the output of the inner function because that output becomes the input of the outer function.

Extension prompt

For the converse, one must also require x_0 to lie in the domain of g ; together, these two conditions are sufficient.

§2 · Limits of Sequences

Basic

Single choice

5 min

Textbook-method adaptation

Q004 · Recognize the Rigorous Definition of a Sequence Limit

Question

Which epsilon-N statement correctly expresses that the sequence $\{a_n\}$ converges to A?

Answer first

A

Knowledge points

- epsilon-N definition of sequence convergence
- order of quantifiers
- control of the tail

Reading the problem and choosing a method

The definition allows N to depend on the requested accuracy ϵ . Once N is chosen, every term with $n > N$ must lie in the ϵ -neighborhood of A.

Step-by-step derivation

Step 1. First, an arbitrary accuracy $\epsilon > 0$ is prescribed; this is the initial universal quantifier.

Step 2. One may then choose a positive integer $N = N(\epsilon)$ depending on that ϵ .

Step 3. The requirement concerns every $n > N$, not merely one selected index: all tail terms satisfy $|a_n - A| < \epsilon$.

Step 4. A has exactly this quantifier order and tail condition. B and C are too strong, while D allows only scattered close terms.

Common pitfalls

- choosing N before ϵ and forcing one N to work for all accuracies
- replacing all sufficiently large indices by the existence of one large index

Verification

The quantifier form of A is $\forall \varepsilon > 0, \exists N \in \mathbb{N}^+, \forall n > N: |a_n - A| < \varepsilon$, which is the standard definition.

Method summary

Remember the order: prescribe ε , choose N , then control every term in the tail.

Extension prompt

Replacing $|a_n - A| < \varepsilon$ by $a_n \in (A - \varepsilon, A + \varepsilon)$ gives an equivalent neighborhood formulation.

§2 · Limits of Sequences

Basic

Fill in the blank

6 min

Textbook-method adaptation

Q005 · Choose N for a Prescribed Accuracy

Question

For $a_n = (-1)^n/n$, the limit is _____. When $\varepsilon = 0.01$ and the definition uses $n > N$, one may take $N =$ _____.

Answer first

0; 100.

Knowledge points

- an alternating sign disappears in an absolute-value estimate
- epsilon-N definition
- strict inequality

Reading the problem and choosing a method

Although $(-1)^n$ alternates in sign, the distance from 0 is always $1/n$. To make this distance strictly smaller than 0.01, it is enough to require $n > 100$.

Step-by-step derivation

Step 1. Compute the distance from the candidate limit 0: $|a_n - 0| = |(-1)^n|/n = 1/n$.

Step 2. The condition $1/n < 0.01 = 1/100$ is equivalent to $n > 100$.

Step 3. Because the definition uses $n > N$, choosing $N = 100$ makes every $n > N$ satisfy the strict inequality.

Step 4. The same reasoning works for every $\varepsilon > 0$ by choosing $N \geq 1/\varepsilon$, so $a_n \rightarrow 0$; the blanks are 0 and 100.

Common pitfalls

- mistaking alternating signs for divergence even when magnitudes go to zero
- thinking N must be 101 even though the convention is $n > N$

Verification

At $n=101$, $|a_{101}|=1/101<0.01$. At $n=100$ the value equals 0.01, but $n=100$ is not part of the tail $n>N$.

Method summary

Convergence controls distance; absolute values remove the alternating sign.

Extension prompt

If the convention were $n\geq N$, one could choose $N=101$ to preserve the strict inequality.

§2 · Limits of Sequences

Basic

True/false with justification

4 min

Q006 · Does Boundedness Guarantee Convergence?

Question

True or false: Every bounded sequence converges.

Answer first

False. The sequence $a_n = (-1)^n$ is bounded but divergent.

Knowledge points

- every convergent sequence is bounded
- the converse is false
- counterexample using even and odd subsequences

Reading the problem and choosing a method

The valid theorem says convergence implies boundedness; it cannot be reversed. A sequence that keeps oscillating between two values gives a direct counterexample.

Step-by-step derivation

Step 1. Let $a_n = (-1)^n$. Then $|a_n| = 1$ for every n , so the sequence is bounded.

Step 2. Its even subsequence satisfies $a_{2n} = 1$ and therefore converges to 1.

Step 3. Its odd subsequence satisfies $a_{2n-1} = -1$ and therefore converges to -1.

Step 4. If the original sequence converged, all subsequences would have the same limit. Since these two limits differ, the original sequence diverges and the statement is false.

Common pitfalls

- reversing a necessary consequence into a sufficient condition
- claiming oscillation without a verifiable argument for divergence

Verification

All terms lie in $[-1, 1]$, confirming boundedness, but no candidate limit can make both positive and negative tail terms arbitrarily close to it.

Method summary

Convergence implies boundedness, but boundedness controls size only and does not prevent perpetual oscillation.

Extension prompt

Adding monotonicity changes the conclusion: every monotone bounded sequence converges, an existence criterion developed later in the chapter.

§3 · Limits of Functions

Basic

Single choice

5 min

Textbook-method adaptation

Q007 · Does a Point Value Affect a Function Limit?

Question

Define $f(1)=7$, and for $x \neq 1$ let $f(x)=(x^2-1)/(x-1)$. What is $\lim_{x \rightarrow 1} f(x)$?

Answer first

C

Knowledge points

- a limit examines a deleted neighborhood
- factor and cancel a common factor
- a point value may differ from the limit

Reading the problem and choosing a method

As $x \rightarrow 1$, only values with x close to but different from 1 are relevant. Therefore the formula for $x \neq 1$ is used, and the assigned value $f(1)=7$ does not enter the limit.

Step-by-step derivation

Step 1. For $x \neq 1$, factor $x^2-1=(x-1)(x+1)$.

Step 2. Within the deleted neighborhood, cancel $x-1$ to obtain $f(x)=x+1$.

Step 3. Therefore, as $x \rightarrow 1$, $f(x)=x+1 \rightarrow 2$.

Step 4. Although $f(1)=7$, changing one point does not alter the nearby trend, so the correct choice is C.

Common pitfalls

- substituting the assigned value $f(1)=7$ merely because x approaches 1
- claiming the original quotient equals $x+1$ at $x=1$, even though cancellation is valid only for $x \neq 1$

Verification

At $x=0.99$ and $x=1.01$, the values are 1.99 and 2.01, both close to 2 rather than 7.

Method summary

A finite-point limit is determined by values in a deleted neighborhood and is independent of the function value at the point itself.

Extension prompt

If $f(1)$ were changed to 2, the function would be continuous at 1. Any other point value would leave the limit equal to 2.

§3 · Limits of Functions

Basic

True/false with justification

4 min

Q008 · Are Two Existing One-Sided Limits Sufficient?

Question

True or false: If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist and are finite, then $\lim_{x \rightarrow a} f(x)$ must exist.

Answer first

False. The two one-sided limits must also be equal.

Knowledge points

- criterion for a two-sided limit
- one-sided limits
- counterexample

Reading the problem and choosing a method

Existence of the two one-sided limits gives a stable trend on each side separately. A single two-sided limit exists only when both trends approach the same value.

Step-by-step derivation

Step 1. A two-sided limit exists exactly when both one-sided limits exist and are equal.

Step 2. The statement assumes existence and finiteness but omits equality.

Step 3. For a counterexample, let $f(x)=0$ for $x < a$ and $f(x)=1$ for $x \geq a$. The left-hand limit is 0 and the right-hand limit is 1.

Step 4. Both one-sided limits exist and are finite, but they are unequal, so the two-sided limit does not exist; the statement is false.

Common pitfalls

- computing both one-sided limits without comparing them
- thinking that the value assigned at a can repair unequal one-sided limits

Verification

Approaching through $x < a$ gives only the value 0, while approaching through $x > a$ gives only 1; no single candidate limit can fit both sides.

Method summary

A two-sided limit exists precisely when the left and right limits exist and equal the same finite value.

Extension prompt

Redefining $f(a)$ cannot change either one-sided limit in this counterexample and therefore cannot create a two-sided limit.

§4 · Infinitesimals and Infinite Quantities

Basic

Single choice

4 min

Q009 · Recognizing an Infinitesimal

Question

As $x \rightarrow 0$, which of the following functions is an infinitesimal?

Answer first

B

Knowledge points

- Definition of an infinitesimal: a function with limit 0
- Term-by-term limit checking
- Difference between boundedness and being infinitesimal

Reading the problem and choosing a method

To decide whether an expression is infinitesimal, check whether its limit is 0 in the specified process. Merely containing x is not enough.

Step-by-step derivation

Step 1. For A, $\lim_{x \rightarrow 0} (2+x) = 2$, so it is not infinitesimal.

Step 2. For B, $0 \leq x^2/(1+|x|) \leq x^2$ and $x^2 \rightarrow 0$; the squeeze theorem gives limit 0.

Step 3. For C, $1/x^2 \rightarrow +\infty$ as $x \rightarrow 0$, so it is infinite rather than infinitesimal.

Step 4. For D, $\lim_{x \rightarrow 0} (1+\sin x) = 1$, so it is not infinitesimal.

Step 5. Therefore B is the unique correct choice.

Common pitfalls

- Calling every expression containing x infinitesimal
- Confusing boundedness with convergence to 0
- Ignoring the specified process $x \rightarrow 0$

Verification

At $x=0.1$ and $x=0.01$, B is about 0.00909 and 0.000099, respectively, while the other expressions approach 2, $+\infty$, and 1.

Method summary

Being infinitesimal is a limit property tied to a specified process, not a visual or absolute-size judgment.

Extension prompt

As $x \rightarrow a$, $f(x)$ is infinitesimal if and only if $\lim_{x \rightarrow a} f(x) = 0$.

§4 · Infinitesimals and Infinite Quantities

Basic

True/false with justification

5 min

Q010 · Is an Infinitesimal a Small Number?

Question

True or false, with justification: An infinitesimal is a nonzero number whose absolute value is very small.

Answer first

False. An infinitesimal is a variable or function whose limit is 0 in a specified process, not a fixed nonzero real number; the constant 0 also qualifies.

Knowledge points

- Process-dependent nature of infinitesimals
- Variables versus constants
- Limits of constant functions

Reading the problem and choosing a method

The statement replaces a limit property by an informal claim about the size of a fixed number. Any fixed positive number, however small, has itself as its constant-function limit.

Step-by-step derivation

Step 1. By definition, $\alpha(x)$ is infinitesimal as $x \rightarrow a$ when $\lim_{x \rightarrow a} \alpha(x) = 0$.

Step 2. For any fixed $c \neq 0$, even if $|c|$ is tiny, the constant function $\alpha(x) = c$ has limit $c \neq 0$.

Step 3. Thus a fixed nonzero small number is not an infinitesimal; the limiting process must be stated.

Step 4. The constant function $\alpha(x) = 0$ has limit 0, so zero itself satisfies the definition.

Step 5. Both parts of the original assertion are therefore inaccurate, so it is false.

Common pitfalls

- Confusing informal smallness with convergence to 0
- Assuming an infinitesimal can never equal 0
- Omitting the limiting process

Verification

For $c=10^{-100}$, $\lim_{(x \rightarrow 0)} c = c \neq 0$; for $\alpha(x)=x$, $\lim_{(x \rightarrow 0)} \alpha(x)=0$. Both can be numerically small, but only the latter tends to 0.

Method summary

An infinitesimal describes a trend of variation, not a fixed magnitude.

Extension prompt

The same function can behave differently in different processes: x is infinitesimal as $x \rightarrow 0$ but not as $x \rightarrow +\infty$.

§5 · Algebra of Limits

Basic

Single choice

4 min

Q011 · When Can Limits Be Divided Directly?

Question

Which conclusion follows directly from the quotient limit law?

Answer first

A

Knowledge points

- Quotient limit law
- Nonzero denominator-limit condition
- Indeterminate forms

Reading the problem and choosing a method

The quotient law requires finite limits for numerator and denominator and a nonzero denominator limit. Only A satisfies all conditions.

Step-by-step derivation

Step 1. In A, $\lim g=3\neq 0$, so the quotient law gives $\lim(f/g)=2/3$.

Step 2. In B, the denominator limit is 0, so a key hypothesis of the law is missing.

Step 3. C is the indeterminate form $0/0$; the two zero limits do not determine the quotient.

Step 4. D is false because if $f(x)=g(x)\neq 0$, their quotient can be identically 1 even while both functions oscillate.

Step 5. Thus A is the unique direct consequence.

Common pitfalls

- Treating $0/0$ as an ordinary fraction
- Forgetting to check that the denominator limit is nonzero
- Assuming two nonexistent limits cannot cancel in a quotient

Verification

A matches every theorem hypothesis. Counterexamples such as $f=x$, $g=x^2$ address B and C, while $f=g=2+\sin(1/x)$ addresses D.

Method summary

Before citing the quotient law, explicitly verify $\lim g \neq 0$.

Extension prompt

The denominator may vanish at isolated points, provided the quotient is defined in a deleted neighborhood and the denominator limit is nonzero.

§5 · Algebra of Limits

Basic

Calculation

4 min

Q012 · Direct Substitution for a Polynomial

Question

Evaluate $\lim_{x \rightarrow 2}(3x^3 - 2x^2 + x - 5)$.

Answer first

13

Knowledge points

- Continuity of polynomials
- Sum, difference, and product limit laws
- Direct substitution

Reading the problem and choosing a method

A polynomial is built from constants and powers by finitely many algebraic operations, so direct substitution is valid at every real number.

Step-by-step derivation

Step 1. From $\lim_{x \rightarrow 2} x = 2$, obtain $\lim_{x \rightarrow 2} x^2 = 4$ and $\lim_{x \rightarrow 2} x^3 = 8$.

Step 2. Multiply by constants: $\lim_{x \rightarrow 2} 3x^3 = 24$, $\lim_{x \rightarrow 2} (-2x^2) = -8$, and $\lim_{x \rightarrow 2} x = 2$.

Step 3. Include the constant term -5 .

Step 4. Combine: $24 - 8 + 2 - 5 = 13$.

Step 5. Therefore the limit is 13.

Common pitfalls

- Dropping a minus sign
- Computing 2^3 as 6
- Overcomplicating a polynomial that permits substitution

Verification

$P(2)=3\cdot 8-2\cdot 4+2-5=13$, matching the limit-law computation.

Method summary

At finite points, substitute directly into polynomials first.

Extension prompt

Every polynomial P satisfies $\lim_{x \rightarrow a} P(x) = P(a)$.

§5 · Algebra of Limits

Basic

Calculation

6 min

Textbook-method
adaptation

Q013 · Removing a Zero Factor with a Difference of Cubes

Question

Evaluate $\lim_{x \rightarrow 2} (x^3 - 8)/(x - 2)$ without derivatives.

Answer first

12

Knowledge points

- Difference-of-cubes identity
- Cancellation in a deleted neighborhood
- Limits versus point values

Reading the problem and choosing a method

Substitution gives $0/0$, which only says the quotient law is not directly applicable. Factor $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ and cancel for $x \neq 2$.

Step-by-step derivation

Step 1. Direct substitution gives $0/0$, signaling that algebraic transformation is needed.

Step 2. Use $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$.

Step 3. For $x \neq 2$, the quotient equals $x^2 + 2x + 4$; cancellation is valid because a limit uses a deleted neighborhood.

Step 4. Substitute $x = 2$ into the polynomial to obtain $4 + 4 + 4 = 12$.

Step 5. Hence the original limit is 12.

Common pitfalls

- Treating $0/0$ as the answer
- Failing to state $x \neq 2$ during cancellation
- Assuming an undefined point prevents a limit

Verification

At $x=2.001$, the quotient equals $x^2+2x+4 \approx 12.006001$, close to 12.

Method summary

The form $0/0$ signals simplification; common zero factors may be canceled in a deleted neighborhood.

Extension prompt

Similarly, $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = na^{n-1}$ follows from factorization, without derivatives.

§6 · Existence Criteria and Two Fundamental Limits

Basic

Single choice

4 min

Q014 · Recognizing the Complete Squeeze-Theorem Hypotheses

Question

Which set of conditions is sufficient to conclude $f(x) \rightarrow L$ by the squeeze theorem?

Answer first

A

Knowledge points

- Squeeze theorem
- Two-sided control
- Common limiting value

Reading the problem and choosing a method

The squeeze theorem needs both bounds to converge to the same number in the same limiting process. One-sided control or different endpoint limits cannot determine f uniquely.

Step-by-step derivation

Step 1. A gives $g \leq f \leq h$ and both bounds tend to L , exactly the squeeze theorem.

Step 2. B gives only a lower bound, so f may remain far above L .

Step 3. C gives only an upper bound, so f may remain far below L .

Step 4. D traps f between two distinct limiting levels but does not force a unique limit.

Step 5. Therefore A is correct.

Common pitfalls

- Using squeeze with only one bound
- Ignoring that the two bound limits must agree
- Failing to ensure the inequalities hold in one neighborhood

Verification

For B take $g=0, f=1$; for C take $h=0, f=-1$; for D take $g=0, h=2, f=1$. None determines the stated L .

Method summary

Squeeze requires two-sided eventual inequalities and a common limit for both bounds.

Extension prompt

The sequence version of the squeeze theorem has the identical logical structure.

§6 · Existence
Criteria and Two
Fundamental Limits

Basic

Calculation

5 min

Textbook-method
adaptation

Q015 · Suppressing Oscillation with a Vanishing Factor

Question

Evaluate $\lim_{x \rightarrow 0} x \sin(1/x)$.

Answer first

0

Knowledge points

- Boundedness of sine
- Absolute-value squeeze
- Oscillation with a vanishing factor

Reading the problem and choosing a method

Although $\sin(1/x)$ oscillates near 0, its absolute value never exceeds 1. Multiplication by x reduces the amplitude to at most $|x|$.

Step-by-step derivation

Step 1. For every $x \neq 0$, $|\sin(1/x)| \leq 1$.

Step 2. Hence $|x \sin(1/x)| = |x| |\sin(1/x)| \leq |x|$.

Step 3. Equivalently, $-|x| \leq x \sin(1/x) \leq |x|$.

Step 4. As $x \rightarrow 0$, both $-|x|$ and $|x|$ tend to 0.

Step 5. The squeeze theorem gives the limit 0.

Common pitfalls

- Trying to find a limit of $\sin(1/x)$ itself
- Saying bounded times zero without a uniform bound
- Forgetting absolute values in the squeeze

Verification

For every x , $|x \sin(1/x)| \leq |x|$; in particular, $|x| < 10^{-3}$ forces the product's absolute value below 10^{-3} .

Method summary

A uniformly bounded oscillatory factor can be controlled by a factor tending to 0.

Extension prompt

If $|g(x)| \leq M$ and $\alpha(x) \rightarrow 0$, then $\alpha(x)g(x) \rightarrow 0$.

§6 · Existence
Criteria and Two
Fundamental Limits

Basic

Fill in the blank

4 min

Textbook-method
adaptation

Q016 · Scaling the First Fundamental Limit

Question

$\lim_{x \rightarrow 0} \sin(5x)/(3x) = \underline{\hspace{1cm}}$.

Answer first

5/3

Knowledge points

- $\lim_{u \rightarrow 0} \sin u/u = 1$
- Creating the standard form
- Constant factors

Reading the problem and choosing a method

Rewrite the denominator using the same argument $5x$ as the sine, preserving the constant ratio.

Step-by-step derivation

Step 1. Write $\sin(5x)/(3x) = (5/3) \cdot [\sin(5x)/(5x)]$.

Step 2. Let $u = 5x$; then $u \rightarrow 0$ as $x \rightarrow 0$.

Step 3. By the first fundamental limit, $\sin u/u \rightarrow 1$.

Step 4. Thus the original limit is $(5/3) \cdot 1 = 5/3$.

Step 5. The blank is $5/3$.

Common pitfalls

- Writing 1 without accounting for scaling
- Dropping the factor $5/3$
- Replacing $\sin(5x)$ by $5\sin x$

Verification

At $x=0.001$, $\sin(0.005)/(0.003)\approx 1.66666$, close to $5/3$.

Method summary

For $\sin(kx)/x$, create $\sin(kx)/(kx)$ and retain the factor k .

Extension prompt

In general, $\lim_{x \rightarrow 0} \sin(ax)/(bx) = a/b$ when $b \neq 0$.

§7 · Comparison of
Infinitesimals

Basic

Single choice

4 min

Q017 · Recognizing Equivalent Infinitesimals

Question

As $x \rightarrow 0$, which equivalence of infinitesimals is correct?

Answer first

A

Knowledge points

- Definition of equivalent infinitesimals
- Standard equivalences
- Comparison of orders

Reading the problem and choosing a method

The test for $\alpha \sim \beta$ is $\alpha/\beta \rightarrow 1$. Apply the fundamental limits to each option.

Step-by-step derivation

Step 1. For A, $\ln(1+x)/x \rightarrow 1$, so $\ln(1+x) \sim x$.

Step 2. For B, $(1-\cos x)/x = [(1-\cos x)/x^2]x \rightarrow (1/2) \cdot 0 = 0$, not 1.

Step 3. For C, $\sin x/x^2 = [\sin x/x] \cdot 1/x$ does not tend to 1.

Step 4. For D, $e^x - 1 \rightarrow 0$ while 1 does not, so they cannot be equivalent infinitesimals.

Step 5. Only A is correct.

Common pitfalls

- Assuming any two infinitesimals are equivalent
- Checking only that both tend to 0
- Ignoring their different orders

Verification

The ratios in A, B, and C tend to 1, 0, and an unbounded quantity, respectively; D does not even compare two infinitesimals.

Method summary

Equivalence means more than both tending to 0: their ratio must tend to 1.

Extension prompt

Other standard relations are $\sin x \sim x$, $\tan x \sim x$, $e^x - 1 \sim x$, and $1 - \cos x \sim x^2/2$.

§8 · Continuity and Discontinuities

Basic

Single choice

4 min

Q018 · The Essential Condition for Continuity

Question

Suppose f is defined in a neighborhood of a , including at a . Which statement is equivalent to saying that f is continuous at a ?

Answer first

A

Knowledge points

- definition of continuity at a point
- relation between a function value and a limit
- two-sided and one-sided limits

Reading the problem and choosing a method

Continuity requires more than the existence of a function value or one one-sided limit. The two-sided limit as x approaches a must exist and must equal the actual value of the function at a .

Step-by-step derivation

Step 1. By definition, f is continuous at a if and only if $\lim_{x \rightarrow a} f(x) = f(a)$.

Step 2. Choice B is insufficient. For example, let $f(0) = 1$ and $f(x) = 0$ for $x \neq 0$; then $f(0)$ exists, but f is not continuous at 0.

Step 3. Choice C supplies only the left-hand limit. Continuity requires both one-sided limits to exist, to be equal, and to have common value $f(a)$.

Step 4. Choice D is also insufficient. The function $\sin(1/x)$ is bounded in every punctured neighborhood of 0, yet no assigned value at 0 makes it continuous there.

Step 5. Thus only choice A states the complete necessary-and-sufficient condition.

Common pitfalls

- confusing being defined with being continuous
- checking only one one-sided limit
- mistaking local boundedness for existence of a limit

Verification

Choice A can be unpacked into three checks: the two-sided limit exists, it is finite, and it equals $f(a)$; these are precisely the requirements in the definition.

Method summary

At a point, always check that the function value exists, the limit exists, and the two are equal.

Extension prompt

At an endpoint of the domain, continuity is usually defined with the appropriate one-sided limit.

§8 · Continuity and Discontinuities

Basic

Fill in the blank

5 min

Textbook-method adaptation

Q019 · Defining a Missing Value to Obtain Continuity

Question

Let $f(x) = (x^2 - 4)/(x - 2)$ for $x \neq 2$, and define $f(2) = c$. To make f continuous at $x = 2$, one must take $c =$.

Answer first

$c = 4$

Knowledge points

- removable discontinuity caused by a common factor
- continuous extension
- assigning the limiting value

Reading the problem and choosing a method

The original quotient is undefined at $x = 2$, but the common factor $x - 2$ can be canceled in a punctured neighborhood. The only value that gives a continuous extension is the limit at that point.

Step-by-step derivation

Step 1. Factor the numerator: $x^2 - 4 = (x - 2)(x + 2)$.

Step 2. For $x \neq 2$, $f(x) = [(x - 2)(x + 2)] / (x - 2) = x + 2$.

Step 3. Hence $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$.

Step 4. Continuity at 2 requires $f(2) = \lim_{x \rightarrow 2} f(x)$.

Step 5. Since $f(2) = c$, the unique choice is $c = 4$.

Common pitfalls

- forgetting that cancellation is valid only for $x \neq 2$
- stopping after direct substitution gives $0/0$
- giving the simplified expression instead of the required numerical value

Verification

With $c=4$, the resulting function agrees with $x+2$ at every real number, so it is continuous at 2 and everywhere else.

Method summary

At a removable discontinuity, define the missing value to be the limit after the removable factor is canceled.

Extension prompt

If $c \neq 4$, then $x=2$ remains a removable discontinuity because the limit exists but differs from the assigned value.

§9 · Operations on Continuous Functions

Basic

Single choice

6 min

Q020 · Finding an Interval of Continuity from the Domain

Question

Which set is both the domain and the interval of continuity of $F(x)=\ln(4-x^2)/\sqrt{x+1}$?

Answer first

B

Knowledge points

- domain of a logarithm
- square-root and denominator restrictions
- continuity of elementary functions on their domains

Reading the problem and choosing a method

A composite elementary function is continuous wherever it is defined. We therefore impose both positivity of the logarithm's argument and strict positivity of the square root that appears in the denominator.

Step-by-step derivation

Step 1. The logarithm requires $4-x^2>0$, which is equivalent to $-2<x<2$.

Step 2. The square root requires $x+1\geq 0$, but because it is in the denominator, zero must also be excluded. Hence $x+1>0$, or $x>-1$.

Step 3. Intersecting the two conditions gives $(-2,2)\cap(-1,+\infty)=(-1,2)$.

Step 4. On this open interval, $4-x^2$, $\ln(4-x^2)$, $\sqrt{x+1}$, and their quotient are continuous by the algebra and composition theorems.

Step 5. Thus both the domain and interval of continuity are $(-1,2)$, so B is correct.

Common pitfalls

- using only $x+1 \geq 0$ and forgetting that a denominator cannot vanish
- replacing the strict logarithm condition by $4-x^2 \geq 0$
- failing to intersect all domain conditions

Verification

At $x=-1$ the denominator is zero, and at $x=2$ the logarithm's argument is zero, so neither endpoint is included. Every $-1 < x < 2$ satisfies both restrictions.

Method summary

The continuity intervals of an elementary expression are the interval components of its domain, found by intersecting every layer of restrictions.

Extension prompt

If the square root were multiplied in the numerator instead, $x=-1$ could be included, subject still to the logarithm condition.

§9 · Operations on
Continuous Functions

Basic

Multiple choice

7 min

Q021 · Algebra Rules for Continuous Functions

Question

Which statements about continuity are correct?

Answer first

A, B, C

Knowledge points

- sums, products, and quotients of continuous functions
- continuity of compositions
- continuity of elementary functions

Reading the problem and choosing a method

A polynomial is obtained from constant and identity functions by finitely many sums and products. A rational function additionally requires a nonzero denominator. A composition requires continuity of both layers and a meaningful expression.

Step-by-step derivation

Step 1. Polynomials are built by finite sums and products of continuous functions, so they are continuous on all of $\mathbb{R}$; A is correct.

Step 2. A rational function is a quotient of polynomials, and the quotient theorem applies wherever its denominator is nonzero; B is correct.

Step 3. If the inner function is continuous at the point, the outer function is continuous at the corresponding image, and the composition is defined, then the composition is continuous; C is correct.

Step 4. The quotient theorem explicitly requires a nonzero denominator at the point. If the denominator is zero, the expression may not even be defined; D is false.

Step 5. Therefore the correct selections are A, B, and C.

Common pitfalls

- omitting the nonzero-denominator condition
- checking only the outer function and not whether the composition is defined
- assuming every elementary expression is continuous on all real numbers

Verification

A direct counterexample to D is $f(x)=1$ and $g(x)=x$: both are continuous at 0, but $f(x)/g(x)=1/x$ is undefined and discontinuous there.

Method summary

Algebra rules for continuity have hypotheses; domain checks are essential for quotients and compositions.

Extension prompt

Compositions involving roots, logarithms, and inverse trigonometric functions must also obey their respective domain restrictions.

§10 · Continuous Functions
on Closed Intervals

Basic

True/false with justification

5 min

Q022 · Are Extrema Guaranteed on an Open Interval?

Question

True or false: If f is continuous on an open interval (a,b) , then it must attain both a maximum and a minimum in (a,b) .

Answer first

False

Knowledge points

- extreme value theorem
- closed-interval hypothesis
- counterexample

Reading the problem and choosing a method

The extreme value theorem requires continuity on a closed interval $[a,b]$. On an open interval, function values may approach endpoint bounds without ever attaining them.

Step-by-step derivation

Step 1. The standard extreme value theorem assumes that f is continuous on the closed interval $[a,b]$.

Step 2. The hypothesis here gives continuity only on (a,b) , so the endpoints are absent and the theorem does not apply.

Step 3. Take the counterexample $f(x)=x$ on $(0,1)$. As a polynomial, it is continuous throughout this open interval.

Step 4. For every $x \in (0,1)$, the point $(x+1)/2 \in (0,1)$ has a larger function value, so no maximum exists.

Step 5. Likewise, $x/2 \in (0,1)$ has a smaller function value, so no minimum exists.

Step 6. Therefore continuity on an open interval does not guarantee attained extrema, and the statement is false.

Common pitfalls

- remembering only that continuous functions have extrema and omitting the closed-interval condition
- confusing a supremum with a maximum
- assuming boundedness guarantees attainment

Verification

For $f(x)=x$ on $(0,1)$, the supremum is 1 and the infimum is 0, but neither endpoint belongs to the domain, so neither bound is attained.

Method summary

Whenever using the extreme value theorem, explicitly check both continuity and the closed-interval hypothesis.

Extension prompt

Even a continuous and bounded function on an open interval may fail to attain either its maximum or its minimum.

Methods

Attempt independently before consulting the solutions.

§1 · Mappings and Functions

Standard

Calculation

8 min

Textbook-method adaptation

Q023 · A Composite Function and Its Domain

Question

Let $f(x) = \sqrt{x+2}$ and $g(u) = 1/(u-1)$. Find the formula and natural domain of $(g \circ f)(x)$.

Answer first

$(g \circ f)(x) = 1/(\sqrt{x+2}-1)$, with domain $[-2, -1) \cup (-1, +\infty)$.

Knowledge points

- order of composition
- the inner function must be defined
- the inner output must lie in the outer domain

Reading the problem and choosing a method

Substitution alone is not enough. The radical must exist, and after substitution the denominator of the outer function must remain nonzero.

Step-by-step derivation

Step 1. The order $g \circ f$ means first compute $f(x) = \sqrt{x+2}$, then use it as the input of g .

Step 2. Substitution gives $(g \circ f)(x) = g(f(x)) = 1/(\sqrt{x+2}-1)$.

Step 3. The inner radical requires $x+2 \geq 0$, so $x \geq -2$.

Step 4. The input of g cannot equal 1, so $\sqrt{x+2} \neq 1$, or $x \neq -1$; combining the restrictions gives $[-2, -1) \cup (-1, +\infty)$.

Common pitfalls

- reversing $g \circ f$ into $f \circ g$
- keeping only $x \geq -2$ and missing the zero denominator at $x = -1$

Verification

At $x=-2$ the denominator is -1 , so the function is defined. At $x=-1$ the denominator is 0 , so the point must be removed. Every other $x \geq -2$ satisfies both restrictions.

Method summary

A composite domain requires two checks: the inner function must be evaluable, and its output must be admissible for the outer function.

Extension prompt

One may also find $f \circ g$; its restrictions differ from those of $g \circ f$, illustrating that composition is generally not commutative.

§1 · Mappings and Functions

Standard

Multiple choice

7 min

Q024 · Analyze Global Properties of a Function

Question

Let $h(x)=|x|/(1+x^2)$ on $\mathbb{R}$. Which statements are correct?

Answer first

A, B

Knowledge points

- test for an even function
- algebraic bounds
- counterexamples to periodicity and monotonicity

Reading the problem and choosing a method

Parity follows by comparing $h(-x)$ and $h(x)$. Boundedness follows from $(|x|-1)^2 \geq 0$. Periodicity and monotonicity can be rejected by consequences or explicit values that contradict their definitions.

Step-by-step derivation

Step 1. $h(-x)=|-x|/(1+(-x)^2)=|x|/(1+x^2)=h(x)$, so A is correct.

Step 2. From $(|x|-1)^2 \geq 0$ we obtain $x^2+1 \geq 2|x|$, hence $0 \leq |x|/(1+x^2) \leq 1/2$; B is correct.

Step 3. As $|x| \rightarrow +\infty$, $h(x) \rightarrow 0$, while $h(1)=1/2$. A nonzero periodic repetition is incompatible with this decay, so C is false.

Step 4. Since $h(1)=1/2$ but $h(2)=2/5 < 1/2$, h is not strictly increasing on $[0, +\infty)$; D is false.

Common pitfalls

- stopping after noticing the absolute value and checking only parity
- asserting boundedness or monotonicity from a sketch without algebraic evidence

Verification

Equality in $x^2+1 \geq 2|x|$ occurs at $|x|=1$, so the upper bound $1/2$ is attained. The values $h(1)=1/2$ and $h(2)=2/5$ directly disprove D.

Method summary

Test each global property independently by its definition or a rigorous inequality.

Extension prompt

The monotonicity of h on $[0, 1]$ and $[1, +\infty)$ can be studied algebraically through differences, without derivatives.

§1 · Mappings and Functions

Standard

Calculation

8 min

Textbook-method adaptation

Q025 · An Inverse Function with an Interval Restriction

Question

The domain of $f(x)=2x-3$ is $[1,4]$. Find the range of f , the formula for its inverse, and the domain and range of the inverse.

Answer first

The range of f is $[-1,5]$. The inverse is $f^{-1}(y)=(y+3)/2$, with domain $[-1,5]$ and range $[1,4]$.

Knowledge points

- a strictly monotone function is invertible on its interval
- the original range becomes the inverse domain
- solve after interchanging input and output roles

Reading the problem and choosing a method

The linear function is strictly increasing on the prescribed interval, so it is one-to-one there. Determine the range from endpoint values, then solve $y=2x-3$ for x .

Step-by-step derivation

Step 1. As x increases from 1 to 4, $2x-3$ increases strictly, so f is one-to-one on $[1,4]$.

Step 2. The endpoint values are $f(1)=-1$ and $f(4)=5$, hence the range of f is $[-1,5]$.

Step 3. Solving $y=2x-3$ gives $x=(y+3)/2$, so $f^{-1}(y)=(y+3)/2$.

Step 4. The inverse swaps domain and range: $\text{Dom}(f^{-1})=[-1,5]$ and $\text{Range}(f^{-1})=[1,4]$.

Common pitfalls

- keeping $[1,4]$ as the domain of the inverse after finding its formula
- confusing $f^{-1}(y)$ with the reciprocal $1/(2y-3)$

Verification

For $x \in [1, 4]$, $f^{-1}(f(x)) = ((2x-3)+3)/2 = x$. For $y \in [-1, 5]$, $f(f^{-1}(y)) = 2 \cdot (y+3)/2 - 3 = y$.

Method summary

Finding an inverse requires swapping not only the variables but also the original domain and range.

Extension prompt

If the original domain were all of $\mathbb{R}$, the same inverse formula would hold, but both its domain and range would be $\mathbb{R}$.

§1 · Mappings and Functions

Standard

Single choice

6 min

Q026 · Find the Least Positive Period of a Sum

Question

What is the least positive period of $p(x)=\sin(2x)+\cos(4x)$?

Answer first

C

Knowledge points

- periods of sine and cosine
- common periods
- least positive period

Reading the problem and choosing a method

The component periods show that π is a period of the sum, but a common period does not by itself prove minimality, and in general one may not assume that every period of a sum is a period of each summand. For a rigorous proof, let $T>0$ be an arbitrary period of the sum and use values at two special points to constrain $\sin(2T)$ and $\cos(2T)$.

Step-by-step derivation

Step 1. For $\sin(2x)$, $2T_1=2\pi$ gives the least positive period $T_1=\pi$.

Step 2. For $\cos(4x)$, $4T_2=2\pi$ gives the least positive period $T_2=\pi/2$.

Step 3. π is a period of both terms, so $p(x+\pi)=p(x)$; hence π is indeed a positive period of p .

Step 4. Let $T>0$ be any period of p . From $p(T)=p(0)=1$, set $s=\sin(2T)$ and use $\cos(4T)=1-2s^2$. Then $s+1-2s^2=1$, so $s\in\{0, 1/2\}$.

Step 5. Also $p(\pi/4)=0$, so $p(\pi/4+T)=0$. With $c=\cos(2T)$, we have $p(\pi/4+T)=c-\cos(4T)=c-(2c^2-1)=0$, hence $c\in\{1, -1/2\}$.

Step 6. Combining these candidates with $s^2+c^2=1$ leaves only $s=0$ and $c=1$. Thus $\sin(2T)=0$ and $\cos(2T)=1$, so $2T=2k\pi$ and $T=k\pi$.

Step 7. Every positive period is therefore a positive integer multiple of π , while π itself is a period. Hence the least positive period is π .

Common pitfalls

- taking the smaller component period automatically
- assuming that every period of a sum must separately be a period of each summand
- ruling out only $\pi/2$ while leaving other positive numbers below π untreated

Verification

Direct substitution verifies $p(x+\pi)=p(x)$. The converse argument proves that every period T must have $T=k\pi$, ruling out not only $\pi/2$ but every other positive number below π .

Method summary

A common period proves only that it is a period; minimality requires a rigorous constraint on an arbitrary candidate period.

Extension prompt

For $\sin(4x)+\cos(4x)$, both components have least positive period $\pi/2$, so the sum also has least positive period $\pi/2$.

§2 · Limits of Sequences

Standard

Calculation

7 min

Textbook-method adaptation

Q027 · Limit of a Rational Sequence with Equal Degrees

Question

Evaluate $\lim_{n \rightarrow \infty} (5n^2 - 3n + 1)/(2n^2 + n - 4)$.

Answer first

5/2.

Knowledge points

- divide by the highest power
- limit laws for sequences
- nonzero denominator limit

Reading the problem and choosing a method

Both numerator and denominator are quadratic. Dividing by n^2 converts all lower-degree terms into multiples of $1/n$ or $1/n^2$, which vanish, leaving the ratio of leading coefficients.

Step-by-step derivation

Step 1. For $n \geq 1$, divide both numerator and denominator by n^2 .

Step 2. The expression becomes $(5 - 3/n + 1/n^2)/(2 + 1/n - 4/n^2)$.

Step 3. As $n \rightarrow \infty$, both $1/n$ and $1/n^2$ tend to 0, so the numerator tends to 5 and the denominator to 2.

Step 4. Since the denominator limit is $2 \neq 0$, the quotient law gives the limit $5/2$.

Common pitfalls

- writing only 'compare leading terms' without showing the valid normalization
- forgetting that the quotient law requires a nonzero denominator limit

Verification

At $n=100$ the ratio is $49701/20096 \approx 2.4732$, already close to 2.5; the lower-order corrections continue to shrink.

Method summary

For equal-degree polynomial ratios at infinity, the limit is the ratio of leading coefficients.

Extension prompt

If the numerator degree were lower than the denominator degree, the same normalization would give limit 0.

§2 · Limits of Sequences

Standard

Proof

11 min

Textbook-method adaptation

Q028 · Prove Convergence of a Rational Sequence from the Definition

Question

Use the epsilon-N definition to prove $\lim_{n \rightarrow \infty} (3n-2)/(n+1) = 3$.

Answer first

Given $\varepsilon > 0$, choose $N = \lceil 5/\varepsilon \rceil$. If $n > N$, then $|(3n-2)/(n+1) - 3| = 5/(n+1) < \varepsilon$, so the limit is 3.

Knowledge points

- structure of an epsilon-N proof
- working backward to choose N
- strict inequality

Reading the problem and choosing a method

First simplify the distance between the sequence and the target 3. Then solve the desired inequality $5/(n+1) < \varepsilon$ for a sufficient lower bound on n .

Step-by-step derivation

Step 1. Let $\varepsilon > 0$ be arbitrary and compute $|(3n-2)/(n+1) - 3|$.

Step 2. Combining the terms gives $|(3n-2-3n-3)/(n+1)| = 5/(n+1)$.

Step 3. To obtain $5/(n+1) < \varepsilon$, it is enough that $n+1 > 5/\varepsilon$. Choose the convenient safe value $N = \lceil 5/\varepsilon \rceil$.

Step 4. If $n > N$, then $n > \lceil 5/\varepsilon \rceil \geq 5/\varepsilon$, so $n+1 > 5/\varepsilon$ and consequently $5/(n+1) < \varepsilon$.

Step 5. Since this works for every $\varepsilon > 0$, the epsilon-N definition gives $(3n-2)/(n+1) \rightarrow 3$.

Common pitfalls

- stating a value of N without verifying that it implies the desired inequality
- allowing N to depend on n instead of only on ε

Verification

For $\varepsilon=0.1$, take $N=50$. If $n>50$, then $5/(n+1) \leq 5/51 < 0.1$, confirming the estimate.

Method summary

A definition proof follows the chain: simplify the error, solve backward for n , choose $N(\varepsilon)$, and verify forward.

Extension prompt

For the general sequence $(a_n+b)/(n+c) \rightarrow a$, simplify the error to $|b-ac|/|n+c|$ and use the same pattern.

§3 · Limits of Functions

Standard

Fill in the blank

7 min

Textbook-method adaptation

Q029 · Determine a Parameter from One-Sided Limits

Question

Let $f(x) = \begin{cases} 2x+a & \text{for } x < 1; \\ x^2+1 & \text{for } x > 1 \end{cases}$. For $\lim_{x \rightarrow 1} f(x)$ to exist, a must equal _____, and the limit is then _____.

Answer first

0; 2.

Knowledge points

- one-sided limits of a piecewise function
- existence criterion for a two-sided limit
- parameter equation

Reading the problem and choosing a method

Use the left piece as $x \rightarrow 1^-$ and the right piece as $x \rightarrow 1^+$. The two-sided limit exists when the two results agree; whether $f(1)$ is defined is irrelevant here.

Step-by-step derivation

Step 1. The left-hand limit is $\lim_{x \rightarrow 1^-} (2x+a) = 2+a$.

Step 2. The right-hand limit is $\lim_{x \rightarrow 1^+} (x^2+1) = 2$.

Step 3. Set the one-sided limits equal: $2+a=2$, which gives $a=0$.

Step 4. With $a=0$, both one-sided limits equal 2, so $\lim_{x \rightarrow 1} f(x) = 2$.

Common pitfalls

- substituting $x=1$ into the wrong branch or demanding that $f(1)$ be specified
- finding a but omitting the common limit value

Verification

When $a=0$, the left expression $2x$ tends to 2 and the right expression x^2+1 tends to 2. If $a \neq 0$, the two limits differ by a .

Method summary

At a piecewise junction, a limit parameter is determined by equality of the left and right limits, independently of the point value.

Extension prompt

If $f(1)=b$ is added and continuity at 1 is required, one must also set $b=2$.

§3 · Limits of Functions

Standard

Calculation

7 min

Textbook-method adaptation

Q030 · Use One-Sided Limits to Show Nonexistence

Question

Evaluate $\lim_{x \rightarrow 0} |x|/x$. If it does not exist, explain why.

Answer first

The limit does not exist; the left-hand limit is -1 and the right-hand limit is 1.

Knowledge points

- piecewise representation of absolute value
- one-sided limits
- criterion for a two-sided limit

Reading the problem and choosing a method

$|x|$ has different formulas on the two sides of 0. Simplify separately on each side and compare the one-sided limits; direct substitution at $x=0$ is not appropriate.

Step-by-step derivation

Step 1. For $x < 0$, $|x| = -x$, so $|x|/x = -1$.

Step 2. Therefore $\lim_{x \rightarrow 0^-} |x|/x = -1$.

Step 3. For $x > 0$, $|x| = x$, so $|x|/x = 1$ and $\lim_{x \rightarrow 0^+} |x|/x = 1$.

Step 4. The one-sided limits -1 and 1 are unequal, so $\lim_{x \rightarrow 0} |x|/x$ does not exist.

Common pitfalls

- canceling $|x|/x$ to 1 and ignoring negative x
- claiming nonexistence only because the denominator is zero at $x=0$, without examining nearby values

Verification

Along $x = 1/n$ the function values are always 1, while along $y = -1/n$ they are always -1. The two approaches give different results.

Method summary

Limits involving absolute values often require sign-based one-sided analysis; unequal one-sided limits rule out a two-sided limit.

Extension prompt

The right-hand limit exists and equals 1, and the left-hand limit exists and equals -1, so the behavior at 0 is a jump.

§3 · Limits of Functions

Standard

Calculation

9 min

Textbook-method adaptation

Q031 · Handle the Sign of a Radical as x Approaches Negative Infinity

Question

Evaluate $\lim_{x \rightarrow -\infty} (2x-1)/\sqrt{x^2+1}$.

Answer first

-2.

Knowledge points

- limits at infinity
- $\sqrt{x^2}=|x|$
- sign on the negative direction

Reading the problem and choosing a method

The crucial identity is $\sqrt{x^2}=|x|$, not x . As $x \rightarrow -\infty$, $|x|=-x$, and this determines the sign of the limit.

Step-by-step derivation

Step 1. Divide numerator and denominator by $|x|$ to obtain $[(2x-1)/|x|]/\sqrt{(1+1/x^2)}$.

Step 2. As $x \rightarrow -\infty$, $x < 0$, so $|x|=-x$ and therefore $x/|x|=-1$.

Step 3. Hence $(2x-1)/|x|=2x/|x|-1/|x| \rightarrow -2-0=-2$.

Step 4. Also $\sqrt{(1+1/x^2)} \rightarrow 1$, so the quotient tends to $-2/1=-2$.

Common pitfalls

- replacing $\sqrt{x^2}$ by x and mishandling the negative direction
- comparing leading coefficients while ignoring the absolute value introduced by the square root

Verification

At $x=-100$, the value is $(-201)/\sqrt{10001}\approx-2.0099$, close to -2 . A result of $+2$ would immediately contradict the numerical sign.

Method summary

Whenever $\sqrt{(x^2)}$ appears in an infinite limit, write $|x|$ first and then use the direction to decide whether $|x|$ equals x or $-x$.

Extension prompt

For the same expression as $x\rightarrow+\infty$, the limit is 2 , showing why the two infinite directions must be treated separately.

§3 · Limits of Functions

Standard

Calculation

6 min

Textbook-method adaptation

Q032 · Cancel the Common Factor Causing Zero Division

Question

Evaluate $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$.

Answer first

6.

Knowledge points

- difference of squares
- cancellation in a deleted neighborhood
- limit versus point value

Reading the problem and choosing a method

Direct substitution gives $0/0$, but the numerator contains the same factor as the denominator. A limit uses x close to 3 with $x \neq 3$, so the common factor can be canceled first.

Step-by-step derivation

Step 1. Factor the numerator using $x^2 - 9 = (x - 3)(x + 3)$.

Step 2. For $x \neq 3$, $(x^2 - 9)/(x - 3) = x + 3$.

Step 3. The deleted neighborhood used by the limit satisfies $x \neq 3$, so this simplification is valid for the limit calculation.

Step 4. Therefore $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3) = \lim_{x \rightarrow 3} (x + 3) = 6$.

Common pitfalls

- treating the direct-substitution form $0/0$ as proof that the limit does not exist
- claiming after cancellation that the original quotient is defined at $x = 3$

Verification

At $x=2.99$ and $x=3.01$, the original quotient equals 5.99 and 6.01, approaching 6 from both sides.

Method summary

The form $0/0$ is not an answer; it signals that algebraic simplification may reveal the limit. A deleted neighborhood permits cancellation of the vanishing factor.

Extension prompt

More generally, $\lim_{x \rightarrow a} (x^2 - a^2)/(x - a) = 2a$ by the same difference-of-squares factorization.

§4 · Infinitesimals
and Infinite Quantities

Standard

Fill in the blank

6 min

Textbook-method
adaptation

Q033 · Writing a Function as a Constant Plus an Infinitesimal

Question

As $x \rightarrow 0$, write $f(x) = (3+2x)/(1+x)$ in the form $f(x) = A + \alpha(x)$, where $\alpha(x)$ is infinitesimal. Find A and $\alpha(x)$.

Answer first

$A=3$ and $\alpha(x) = -x/(1+x)$.

Knowledge points

- Equivalence between $\lim f=A$ and $f=A+\alpha$
- Algebraic decomposition
- Verification of an infinitesimal

Reading the problem and choosing a method

First obtain the candidate limit A by substitution, then construct the exact remainder $\alpha=f-A$ and verify that it tends to 0.

Step-by-step derivation

Step 1. Since the denominator at $x=0$ is $1 \neq 0$, $A = \lim_{x \rightarrow 0} (3+2x)/(1+x) = 3$.

Step 2. Define the remainder by $\alpha(x) = f(x) - A$.

Step 3. Compute $\alpha(x) = (3+2x)/(1+x) - 3 = (3+2x-3-3x)/(1+x) = -x/(1+x)$.

Step 4. As $x \rightarrow 0$, the numerator $-x \rightarrow 0$ and the denominator $1+x \rightarrow 1 \neq 0$, so $\alpha(x) \rightarrow 0$.

Step 5. Hence the required decomposition is $f(x) = 3 - x/(1+x)$.

Common pitfalls

- Giving A without verifying that α is infinitesimal
- Expanding $3(1+x)$ incorrectly
- Replacing an exact decomposition by an approximation

Verification

Combining $3-x/(1+x)$ gives $[3(1+x)-x]/(1+x)=(3+2x)/(1+x)$, exactly the original function.

Method summary

When $\lim f=A$, the natural exact remainder is $\alpha=f-A$, and it must be infinitesimal.

Extension prompt

Conversely, $f=A+\alpha$ with $\alpha \rightarrow 0$ immediately implies $\lim f=A$.

§4 · Infinitesimals
and Infinite Quantities

Standard

Calculation

7 min

Textbook-method
adaptation

Q034 · Infinite Behavior of a Reciprocal Square

Question

Find $\lim_{x \rightarrow 0} 1/x^2$, and explain why the reciprocal relationship between $1/x^2$ and x^2 is valid as $x \rightarrow 0$.

Answer first

$\lim_{x \rightarrow 0} 1/x^2 = +\infty$. Since $x^2 \rightarrow 0$ and $x^2 > 0$ in a deleted neighborhood of 0, its reciprocal tends to $+\infty$.

Knowledge points

- Definition of a positive infinite limit
- Reciprocal relation between infinitesimals and infinite quantities
- Deleted neighborhoods

Reading the problem and choosing a method

Besides stating $+\infty$, one must check that the infinitesimal denominator is nonzero near the point. Squaring also guarantees the same sign from both sides.

Step-by-step derivation

Step 1. As $x \rightarrow 0$, $x^2 \rightarrow 0$, and $x^2 > 0$ whenever $x \neq 0$.

Step 2. Given any $M > 0$, choose $\delta = 1/\sqrt{M}$.

Step 3. If $0 < |x| < \delta$, then $x^2 < \delta^2 = 1/M$, hence $1/x^2 > M$.

Step 4. This is exactly the definition of $\lim_{x \rightarrow 0} 1/x^2 = +\infty$.

Step 5. Thus x^2 is an eventually positive nonzero infinitesimal, and its reciprocal $1/x^2$ is positive infinite.

Common pitfalls

- Assigning opposite signs from the two sides despite the square
- Taking a reciprocal without checking nonvanishing
- Treating $+\infty$ as an ordinary real number

Verification

For $M=10^4$, choose $\delta=0.01$. Every x with $0<|x|<0.01$ satisfies $1/x^2>10^4$, exactly as required.

Method summary

An infinite limit should be verifiable against every threshold M ; the square forces $+\infty$ from both sides.

Extension prompt

If $\alpha(x)\rightarrow 0$ and $\alpha(x)\neq 0$ in a deleted neighborhood, then $|1/\alpha(x)|\rightarrow\infty$; its sign determines $+\infty$, $-\infty$, or only divergence in absolute value.

§5 · Algebra of Limits

Standard

Calculation

7 min

Textbook-method
adaptation

Q035 · Rationalizing with a Conjugate

Question

Evaluate $\lim_{x \rightarrow 1} (\sqrt{x+3}-2)/(x-1)$ without derivatives.

Answer first

$1/4$

Knowledge points

- Conjugate rationalization
- Cancellation in a deleted neighborhood
- Domains of radicals

Reading the problem and choosing a method

Both numerator and denominator tend to 0. Multiply by the conjugate to turn the radical difference into $x-1$, then cancel.

Step-by-step derivation

Step 1. Direct substitution gives $(2-2)/(1-1)=0/0$.

Step 2. Multiply numerator and denominator by $\sqrt{x+3}+2$.

Step 3. The numerator becomes $(x+3)-4=x-1$, so for $x \neq 1$ the expression equals $1/(\sqrt{x+3}+2)$.

Step 4. The simplified denominator tends to $2+2=4$ as $x \rightarrow 1$.

Step 5. Therefore the limit is $1/4$.

Common pitfalls

- Expanding the conjugate product incorrectly
- Returning to the unsimplified $0/0$ expression
- Ignoring the local radical domain

Verification

At $x=1.01$, the simplified expression is $1/(\sqrt{4.01}+2)\approx 0.249844$, close to 0.25 .

Method summary

When a difference of radicals creates $0/0$, try conjugate rationalization first.

Extension prompt

In general, the conjugate of $\sqrt{u}-\sqrt{v}$ is $\sqrt{u}+\sqrt{v}$.

§5 · Algebra of Limits

Standard

Calculation

8 min

Textbook-method
adaptation

Q036 · Algebraic Simplification of a Cube-Root Difference

Question

Evaluate $\lim_{x \rightarrow 8} (\sqrt[3]{x} - 2)/(x - 8)$ without derivatives.

Answer first

1/12

Knowledge points

- Factorization of $a^3 - b^3$
- Cube-root substitution
- Cancellation in a deleted neighborhood

Reading the problem and choosing a method

Let $u = \sqrt[3]{x}$. Then $x - 8 = u^3 - 2^3$ factors by $u - 2$, exactly the numerator factor to cancel.

Step-by-step derivation

Step 1. Direct substitution gives 0/0. Set $u = \sqrt[3]{x}$, so $u \rightarrow 2$ as $x \rightarrow 8$.Step 2. Factor $x - 8 = u^3 - 8 = (u - 2)(u^2 + 2u + 4)$.Step 3. For $x \neq 8$, equivalently $u \neq 2$, the expression equals $1/(u^2 + 2u + 4)$.Step 4. As $u \rightarrow 2$, the denominator tends to $4 + 4 + 4 = 12$.

Step 5. Therefore the limit is 1/12.

Common pitfalls

- Using a square-root conjugate formula
- Omitting the middle term $2u$
- Failing to state that $u \rightarrow 2$

Verification

At $u=2.001$, the simplified value is about $1/12.012001$, close to $1/12$.

Method summary

A cube-root difference pairs with the difference-of-cubes identity, not a square conjugate.

Extension prompt

For n th roots, use $a^n - b^n = (a - b)(a^{n-1} + \cdots + b^{n-1})$.

§5 · Algebra of Limits

Standard

Fill in the blank

6 min

Q037 · A Rational Expression from Known Limits

Question

If $f(x) \rightarrow 2$ and $g(x) \rightarrow -1$ as $x \rightarrow a$, then $\lim_{x \rightarrow a} [3f(x) - g(x)] / [f(x) - g(x)] = \underline{\hspace{2cm}}$.

Answer first

7/3

Knowledge points

- Limits of linear combinations
- Quotient limit law
- Nonzero denominator check

Reading the problem and choosing a method

Find the numerator and denominator limits separately, then verify that the denominator limit is nonzero before dividing.

Step-by-step derivation

Step 1. The numerator limit is $3 \cdot 2 - (-1) = 7$.

Step 2. The denominator limit is $2 - (-1) = 3$.

Step 3. Because the denominator limit $3 \neq 0$, the quotient law applies.

Step 4. Thus the requested limit is $7/3$.

Step 5. The blank is $7/3$.

Common pitfalls

- Mishandling the sign of $-g$
- Not checking the denominator limit
- Confusing $3f$ with f^3

Verification

Temporarily replacing f by 2 and g by -1 gives $[6+1]/[2+1]=7/3$, matching the result.

Method summary

For nested limit operations, use the sequence: numerator, denominator, nonzero check.

Extension prompt

The same procedure works for any rational combination whose denominator limit is nonzero.

§5 · Algebra of Limits

Standard

True/false with justification

7 min

Q038 · A Quotient of Two Infinitesimals

Question

True or false, with justification: If $\alpha(x) \rightarrow 0$ and $\beta(x) \rightarrow 0$, then $\alpha(x)/\beta(x) \rightarrow 1$.

Answer first

False. Take $\alpha(x)=x$ and $\beta(x)=x^2$ as $x \rightarrow 0$. Both tend to 0, but $\alpha/\beta=1/x$ has no finite two-sided limit.

Knowledge points

- The indeterminate form $0/0$
- Counterexample method
- Two-sided limits

Reading the problem and choosing a method

The facts that both quantities tend to 0 describe their individual sizes but do not determine the ratio of their rates of decay.

Step-by-step derivation

Step 1. Let $\alpha(x)=x$ and $\beta(x)=x^2$, with $x \rightarrow 0$.

Step 2. Clearly $\lim \alpha=0$ and $\lim \beta=0$, so every hypothesis is satisfied.

Step 3. For $x \neq 0$, $\alpha/\beta=x/x^2=1/x$.

Step 4. As $x \rightarrow 0^+$, $1/x \rightarrow +\infty$, while as $x \rightarrow 0^-$, $1/x \rightarrow -\infty$.

Step 5. The two-sided limit therefore does not exist and is certainly not 1.

Common pitfalls

- Canceling $0/0$ to get 1
- Checking only the premises without the quotient
- Ignoring opposite one-sided behavior

Verification

At $x=0.01$ the quotient is 100, while at $x=-0.01$ it is -100 , so it is not approaching 1.

Method summary

The form $0/0$ is indeterminate; compare orders or simplify algebraically.

Extension prompt

With the additional condition $\alpha(x) \sim \beta(x)$, the definition is precisely $\alpha(x)/\beta(x) \rightarrow 1$.

§5 · Algebra of Limits

Standard

Multiple choice

7 min

Textbook-method
adaptation

Q039 · A Rational Limit at Negative Infinity

Question

For $L = \lim_{x \rightarrow -\infty} (3x^2 - x + 1)/(2x^2 + 5)$, which statements are correct? Select all that apply.

Answer first

A, B, and C

Knowledge points

- Limits at infinity
- Division by the highest power
- Algebraic limit laws

Reading the problem and choosing a method

Both numerator and denominator are quadratic. Dividing by x^2 turns every lower-order term into a quantity tending to 0.

Step-by-step derivation

Step 1. As $x \rightarrow -\infty$, $x \neq 0$, so divide numerator and denominator by x^2 .

Step 2. The expression becomes $(3 - 1/x + 1/x^2)/(2 + 5/x^2)$.

Step 3. Both $1/x \rightarrow 0$ and $1/x^2 \rightarrow 0$.

Step 4. The numerator limit is 3 and the denominator limit is $2 \neq 0$.

Step 5. The quotient law gives $L = 3/2$, so A is true. The same analysis holds for $x \rightarrow +\infty$, so B is true; C is the valid simplification, while D is false because a square is nonnegative.

Common pitfalls

- Treating x^2 as negative because $x \rightarrow -\infty$
- Dividing only some terms
- Forgetting the nonzero denominator-limit check

Verification

At $x=-100$ the value is $30101/20005 \approx 1.50467$, close to 1.5.

Method summary

For equal-degree rational functions at infinity, the limit is the ratio of leading coefficients.

Extension prompt

If the numerator degree is lower, the limit is 0; if higher, analyze the sign of the dominant power.

§6 · Existence
Criteria and Two
Fundamental Limits

Standard

Calculation

7 min

Textbook-method
adaptation

Q040 · Standardizing Two Sine Factors

Question

Evaluate $\lim_{x \rightarrow 0} [\sin(2x)\sin(3x)]/x^2$.

Answer first

6

Knowledge points

- First fundamental limit
- Product limit theorem
- Scale factors

Reading the problem and choosing a method

Pair each sine with its own argument and distribute x^2 into $(2x)(3x)$, retaining the factor 6.

Step-by-step derivation

Step 1. Rewrite the expression as $6 \cdot [\sin(2x)/(2x)] \cdot [\sin(3x)/(3x)]$.

Step 2. As $x \rightarrow 0$, both $2x \rightarrow 0$ and $3x \rightarrow 0$.

Step 3. The first fundamental limit makes each bracket tend to 1.

Step 4. By the product law, the total limit is $6 \cdot 1 \cdot 1$.

Step 5. Therefore the answer is 6.

Common pitfalls

- Pairing x^2 with only one sine
- Dropping $2 \cdot 3 = 6$
- Using the false identity $\sin(2x)\sin(3x) = \sin(6x)$

Verification

At $x=0.001$ the expression is about 5.999987, close to 6.

Method summary

For products of small-angle sines, create $\sin u/u$ factor by factor and preserve every scale constant.

Extension prompt

$$\lim_{x \rightarrow 0} \prod_{k=1}^m \sin(a_k x) / x^m = \prod_{k=1}^m a_k.$$

§6 · Existence
Criteria and Two
Fundamental Limits

Standard

Calculation

8 min

Textbook-method
adaptation

Q041 · Deriving a Cosine Limit from the First Fundamental Limit

Question

Evaluate $\lim_{x \rightarrow 0} [1 - \cos(2x)]/x^2$ without Taylor expansions.

Answer first

2

Knowledge points

- Identity $1 - \cos 2x = 2\sin^2 x$
- First fundamental limit
- Limits of squares

Reading the problem and choosing a method

Convert the cosine difference into a sine square so that $\sin x/x \rightarrow 1$ applies directly.

Step-by-step derivation

Step 1. Use $1 - \cos(2x) = 2\sin^2 x$.

Step 2. The expression becomes $2\sin^2 x/x^2 = 2[\sin x/x]^2$.

Step 3. As $x \rightarrow 0$, $\sin x/x \rightarrow 1$.

Step 4. Therefore $[\sin x/x]^2 \rightarrow 1$.

Step 5. The requested limit is $2 \cdot 1 = 2$.

Common pitfalls

- Writing $1 - \cos 2x = \sin^2 x$
- Losing the square in the denominator
- Using Taylor expansions in a Chapter 1 solution

Verification

At $x=0.01$, $[1 - \cos 0.02]/0.0001 \approx 1.99993$, close to 2.

Method summary

For a cosine difference, first use a half-angle identity to reach the sine fundamental limit.

Extension prompt

Similarly, $\lim_{x \rightarrow 0} [1 - \cos(ax)]/x^2 = a^2/2$.

§6 · Existence
Criteria and Two
Fundamental Limits

Standard

Calculation

8 min

Textbook-method
adaptation

Q042 · Creating the Second Fundamental Limit

Question

Evaluate $\lim_{x \rightarrow 0} (1+2x)^{3/x}$.

Answer first

e^6

Knowledge points

- $\lim_{u \rightarrow 0} (1+u)^{1/u} = e$
- Exponent decomposition
- Composite limits

Reading the problem and choosing a method

Set $u=2x$ and rewrite $3/x$ as $6/u$, producing the sixth power of the standard fundamental-limit expression.

Step-by-step derivation

Step 1. Let $u=2x$; then $u \rightarrow 0$ as $x \rightarrow 0$.

Step 2. The exponent is $3/x = 3/(u/2) = 6/u$.

Step 3. Thus $(1+2x)^{3/x} = [(1+u)^{1/u}]^6$.

Step 4. By the second fundamental limit, $(1+u)^{1/u} \rightarrow e$.

Step 5. Hence the original limit is e^6 .

Common pitfalls

- Computing the exponent scale as $3/2$
- Concluding 1 merely because the base tends to 1
- Ignoring positivity of the base near 0

Verification

At $x=0.001$, $1.002^{(3000)} \approx e^{(5.994)}$, close to e^6 .

Method summary

For a 1^∞ form, standardize both the base increment and exponent; their product controls the final exponent.

Extension prompt

In general, $\lim_{(x \rightarrow 0)} (1+ax)^{(b/x)} = e^{(ab)}$.

§6 · Existence
Criteria and Two
Fundamental Limits

Standard

Calculation

8 min

Textbook-method
adaptation

Q043 · An Exponential Sequence with a Negative Increment

Question

Evaluate $\lim_{n \rightarrow \infty} (1 - 3/n)^n$.

Answer first

e^{-3}

Knowledge points

- Sequence form of the second fundamental limit
- Negative approach to zero
- Exponent scaling

Reading the problem and choosing a method

Let $u_n = -3/n$, so $u_n \rightarrow 0$ and $n = -3/u_n$. The base is positive for all sufficiently large n .

Step-by-step derivation

Step 1. For $n > 3$, $1 - 3/n > 0$, so the real power is well defined.

Step 2. Set $u_n = -3/n$; then $u_n \rightarrow 0$.

Step 3. Since $n = -3/u_n$, $(1 - 3/n)^n = [(1 + u_n)^{1/u_n}]^{-3}$.

Step 4. The second fundamental limit gives $(1 + u_n)^{1/u_n} \rightarrow e$.

Step 5. Therefore the sequence limit is e^{-3} .

Common pitfalls

- Dropping the minus sign and obtaining e^3
- Letting finitely many early terms affect the tail limit
- Treating 1^n as an ordinary power

Verification

At $n=1000$, $(0.997)^{(1000)} \approx 0.04956$, while $e^{-3} \approx 0.04979$.

Method summary

The second fundamental limit also permits a negative increment; the increment-exponent product remains decisive.

Extension prompt

In general, $\lim_{n \rightarrow \infty} (1+c/n)^n = e^c$.

§6 · Existence Criteria and Two Fundamental Limits

Standard

Multiple choice

7 min

Q044 · Recognizing Common Indeterminate Forms

Question

Which limit structures are indeterminate forms? Select all that apply.

Answer first

A, B, and C

Knowledge points

- Indeterminate-form types
- Limit structure
- Second fundamental limit

Reading the problem and choosing a method

An indeterminate form means the component limits alone do not determine the whole. The forms $0/0$, ∞/∞ , and 1^∞ can lead to many outcomes; 0^1 normally tends to 0 under the usual real-domain conditions.

Step-by-step derivation

Step 1. A can yield 0, a finite nonzero value, infinity, or no limit, so $0/0$ is indeterminate.

Step 2. B depends on relative growth rates, so ∞/∞ is indeterminate.

Step 3. C depends on the product of the base's deviation from 1 and the exponent; it is indeterminate.

Step 4. D has a base tending to 0 and an exponent tending to 1, with no 0^0 -type competition, so it normally tends to 0.

Step 5. Thus select A, B, and C.

Common pitfalls

- Treating symbols involving ∞ as arithmetic numbers
- Saying every power with base near 1 tends to 1
- Confusing 0^1 with 0^0

Verification

As $x \rightarrow 0$, compare x/x with x^2/x for $0/0$; as $x \rightarrow +\infty$, compare x/x with x^2/x for ∞/∞ ; and compare $(1+1/n)^n$ with $(1+1/n)^{2n}$ for 1^∞ . Each form can produce different limits.

Method summary

An indeterminate form is a structural warning, not an answer; further transformation is required.

Extension prompt

Other common forms in this chapter include $0 \cdot \infty$, $\infty - \infty$, and 0^0 .

§7 · Comparison of
Infinitesimals

Standard

Fill in the blank

5 min

Q045 · Comparing Orders of Power Infinitesimals

Question

As $x \rightarrow 0$, $|x|^3$ is a ____-order infinitesimal relative to x^2 because $|x|^3/x^2 \rightarrow$ ____.

Answer first

higher; 0.

Knowledge points

- Definition of higher-order infinitesimals
- Absolute-value powers
- Order comparison by a ratio

Reading the problem and choosing a method

If $\alpha/\beta \rightarrow 0$, then α is of higher order than β . Here the ratio simplifies directly to $|x|$.

Step-by-step derivation

Step 1. Let $\alpha(x) = |x|^3$ and $\beta(x) = x^2 = |x|^2$.

Step 2. For $x \neq 0$, $\alpha/\beta = |x|^3/|x|^2 = |x|$.

Step 3. As $x \rightarrow 0$, $|x| \rightarrow 0$.

Step 4. Thus $\alpha/\beta \rightarrow 0$, so α is of higher order than β .

Step 5. The blanks are higher and 0.

Common pitfalls

- Calling a larger exponent lower order
- Ignoring $x^2 = |x|^2$
- Thinking higher order means larger numerical size

Verification

At $x=0.1$ the ratio is 0.1, and at $x=0.01$ it is 0.01, clearly approaching 0.

Method summary

Near 0, a larger positive power usually vanishes faster and is therefore of higher order.

Extension prompt

If $p>q>0$, then $|x|^p$ is higher order than $|x|^q$.

§7 · Comparison of
Infinitesimals

Standard

Calculation

8 min

Textbook-method
adaptation

Q046 · Equivalent Substitution in a Product-Quotient Structure

Question

Evaluate $\lim_{x \rightarrow 0} [\sin(3x) \cdot \ln(1+2x)] / [x(e^x - 1)]$.

Answer first

6

Knowledge points

- $\sin u \sim u$
- $\ln(1+u) \sim u$
- $e^u - 1 \sim u$
- Equivalent substitution in products and quotients

Reading the problem and choosing a method

Every substituted infinitesimal appears as a multiplicative or divisive factor, so the replacements are safe.

Step-by-step derivation

Step 1. As $x \rightarrow 0$, $\sin(3x) \sim 3x$.Step 2. Also $\ln(1+2x) \sim 2x$ and $e^x - 1 \sim x$.Step 3. After factorwise substitution, the expression is equivalent to $[(3x)(2x)]/[x \cdot x]$.

Step 4. This simplifies to the constant 6.

Step 5. Therefore the limit is 6.

Common pitfalls

- Replacing $\ln(1+2x)$ by x instead of $2x$
- Dropping the existing x in the denominator
- Using equivalent substitution without checking the algebraic position

Verification

Alternatively factor it as $[\sin 3x/(3x)] \cdot [\ln(1+2x)/(2x)] \cdot [x/(e^x-1)] \cdot 6$; the first three factors tend to 1.

Method summary

In products and quotients, apply equivalences factor by factor and preserve every scale constant.

Extension prompt

A factor not tending to 0 can simply be handled by its ordinary limit rather than forced into an infinitesimal equivalence.

§7 · Comparison of
Infinitesimals

Standard

Calculation

8 min

Textbook-method
adaptation

Q047 · Balancing Second- and First-Order Factors

Question

Evaluate $\lim_{x \rightarrow 0} [1 - \cos(2x)] / [x \sin(3x)]$.

Answer first

 $2/3$

Knowledge points

- $1 - \cos u \sim u^2/2$
- $\sin u \sim u$
- Equivalent infinitesimals

Reading the problem and choosing a method

The numerator is second order, and the denominator is a product of two first-order factors, also second order.
The remaining constants determine the limit.

Step-by-step derivation

Step 1. As $x \rightarrow 0$, $1 - \cos(2x) \sim (2x)^2/2 = 2x^2$.Step 2. Also $\sin(3x) \sim 3x$.Step 3. Hence $x \sin(3x) \sim x \cdot 3x = 3x^2$.Step 4. The quotient is equivalent to $2x^2/(3x^2) = 2/3$.Step 5. Therefore the limit is $2/3$.

Common pitfalls

- Replacing $1-\cos u$ by u
- Dropping the factor $1/2$
- Forgetting the extra x in the denominator

Verification

Using $1-\cos 2x=2\sin^2 x$ gives $2[\sin x/x]^2 \cdot [x/\sin 3x]$, with the same limit $2/3$.

Method summary

Match the total orders first, then collect the constants left by the equivalences.

Extension prompt

$\lim_{x \rightarrow 0} [1-\cos(ax)]/[x \sin(bx)] = a^2/(2b)$ for $b \neq 0$.

§8 · Continuity and Discontinuities

Standard

True/false with justification

5 min

Q048 · Can a Removable Discontinuity Always Be Repaired?

Question

True or false: If $x=a$ is a removable discontinuity of f , then redefining only $f(a)$ can always make f continuous at a .

Answer first

True

Knowledge points

- definition of a removable discontinuity
- finite limit
- uniqueness of a continuous extension

Reading the problem and choosing a method

At a removable discontinuity, the finite two-sided limit already exists. The only defect is that the function is undefined at the point or has a value different from that limit.

Step-by-step derivation

Step 1. By definition of a removable discontinuity, there is a finite number L such that $\lim_{x \rightarrow a} f(x) = L$.

Step 2. The original $f(a)$ is either undefined or is defined with $f(a) \neq L$.

Step 3. Define a new function F by $F(x) = f(x)$ for $x \neq a$ and $F(a) = L$.

Step 4. Changing one point does not change a punctured limit, so $\lim_{x \rightarrow a} F(x) = L$.

Step 5. Since $F(a) = L$, we have $\lim_{x \rightarrow a} F(x) = F(a)$, and therefore F is continuous at a .

Common pitfalls

- trying to repair a jump discontinuity by changing one point
- ignoring the requirement that the one-sided limits agree
- thinking that any newly assigned value will work

Verification

The repaired function satisfies all three continuity checks: $F(a)$ exists, the two-sided limit exists, and both equal L .

Method summary

The word removable means that the point can be repaired by assigning the unique limiting value.

Extension prompt

If the two one-sided limits are finite but unequal, the point is a jump discontinuity and no single assigned value can repair it.

§8 · Continuity and Discontinuities

Standard

Calculation

12 min

Textbook-method adaptation

Q049 · Parameters for Continuity at Two Junctions

Question

Let $f(x)=ax+b$ for $x<1$, $f(x)=x^2+1$ for $1\leq x\leq 2$, and $f(x)=3x-a$ for $x>2$. Find a, b so that f is continuous on $\mathbb{R}$.

Answer first

$a=1, b=1$

Knowledge points

- continuity of a piecewise function
- one-sided limits at junctions
- linear system

Reading the problem and choosing a method

Each formula is a polynomial and is therefore continuous inside its own interval. Only the junctions $x=1$ and $x=2$ can cause a loss of continuity, so we impose a matching condition at each one.

Step-by-step derivation

Step 1. On $(-\infty, 1)$, $(1, 2)$, and $(2, +\infty)$, each piece is a polynomial and needs no further condition.

Step 2. At $x=1$, the left-hand limit is $a+b$, while the function value and right-hand limit are $1^2+1=2$; hence $a+b=2$.

Step 3. At $x=2$, the left-hand limit and the function value are $2^2+1=5$.

Step 4. The right-hand limit at $x=2$ is $3\cdot 2-a=6-a$, so continuity requires $6-a=5$.

Step 5. The second equation gives $a=1$; substituting into $a+b=2$ gives $b=1$.

Step 6. Thus the unique parameters making both junctions continuous are $a=1, b=1$.

Common pitfalls

- checking only one junction
- using the left piece to compute the value at $x=1$ even though it excludes that point
- matching one-sided limits without checking the actual function value

Verification

With these values, the left-hand limit at 1 is $1+1=2=f(1)$, and the right-hand limit at 2 is $6-1=5=f(2)$. Every other point lies inside a continuous polynomial piece.

Method summary

For a piecewise function, first reduce the work to the points where the defining formula changes.

Extension prompt

If continuity were required only at $x=1$, the parameters would satisfy the line $a+b=2$; the second junction makes the pair unique.

§8 · Continuity and Discontinuities

Standard

Multiple choice

8 min

Q050 · Distinguishing Types of Discontinuity

Question

Which statements about a discontinuity at $x=a$ are correct?

Answer first

A, B, C

Knowledge points

- removable discontinuity
- jump discontinuity
- infinite discontinuity
- oscillatory discontinuity

Reading the problem and choosing a method

The main dividing line is whether both one-sided limits exist as finite numbers. If they do, the discontinuity is of the first kind; if at least one finite one-sided limit fails to exist, it is of the second kind.

Step-by-step derivation

Step 1. In A, the equal finite one-sided limits give a finite two-sided limit, and only the point value is defective; thus A correctly describes a removable discontinuity.

Step 2. In B, the two finite one-sided limits are unequal, producing a finite jump; thus B is correct.

Step 3. In C, an infinite one-sided limit is not a finite limit, so the point is an infinite discontinuity of the second kind; thus C is correct.

Step 4. In D, persistent oscillation prevents at least one one-sided limit from existing, so the point is of the second kind, not the first; thus D is false.

Step 5. The correct selections are therefore A, B, and C.

Common pitfalls

- interpreting first kind as the more severe kind
- classifying two infinite sides as first kind
- forgetting that both infinite and oscillatory discontinuities are of the second kind

Verification

Apply one uniform test: A and B have two finite one-sided limits, whereas C and D lose at least one finite one-sided limit. The classifications are therefore consistent.

Method summary

First determine whether both finite one-sided limits exist; only then distinguish removable, jump, infinite, and oscillatory cases.

Extension prompt

A second-kind discontinuity can be bounded: $\sin(1/x)$ near $x=0$ is the standard oscillatory example.

§8 · Continuity and Discontinuities

Standard

Proof

14 min

Textbook-method adaptation

Q051 · Proving Continuity of a Quadratic from the Definition

Question

Without citing the theorem that polynomials are continuous, use the ε - δ definition to prove that $f(x)=x^2+1$ is continuous at every $a \in \mathbb{R}$.

Answer first

For any $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/(2|a|+1)\}$; then $|(x^2+1)-(a^2+1)| < \varepsilon$ follows.

Knowledge points

- epsilon-delta definition of continuity
- difference of squares
- auxiliary restriction

Reading the problem and choosing a method

The target difference factors as $|x-a||x+a|$. The first factor is controlled by $|x-a| < \delta$, while the second still contains the variable x . We first impose $|x-a| < 1$ to bound it by a constant depending only on the fixed point a .

Step-by-step derivation

Step 1. Let $\varepsilon > 0$ be arbitrary. We seek $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x)-f(a)| < \varepsilon$.

Step 2. Factor the target difference: $|f(x)-f(a)| = |x^2-a^2| = |x-a||x+a|$.

Step 3. If $|x-a| < 1$, then $|x| \leq |x-a| + |a| < 1 + |a|$, so $|x+a| \leq |x| + |a| < 2|a| + 1$.

Step 4. Choose $\delta = \min\{1, \varepsilon/(2|a|+1)\}$. This is positive and enforces both the auxiliary bound and the desired accuracy.

Step 5. Whenever $|x-a| < \delta$, we have $|f(x)-f(a)| < |x-a|(2|a|+1) < \delta(2|a|+1) \leq \varepsilon$.

Step 6. Thus the definition is satisfied at a . Since a was arbitrary, $f(x)=x^2+1$ is continuous on $\mathbb{R}$.

Common pitfalls

- setting $\delta = \varepsilon/|x+a|$, which still depends on the variable x
- forgetting that a is fixed while x varies
- failing to control $|x-a|$ before bounding the second factor

Verification

The chosen δ depends only on the given ε and the fixed point a , not on x , and the final estimate is strictly below ε in the required order of quantifiers.

Method summary

When a product contains an uncontrolled variable factor, first use a simple neighborhood restriction to bound that factor by a constant.

Extension prompt

The same factorization-and-bounding strategy can be iterated to prove continuity of every power x^n and hence every polynomial.

§9 · Operations on Continuous Functions

Standard

Calculation

9 min

Textbook-method adaptation

Q052 · Evaluating a Limit by Continuity of Elementary Functions

Question

Use continuity to evaluate $\lim_{x \rightarrow \pi/6} (\ln(2+\cos x))/(\sqrt{1+\sin x})$, and explain why direct substitution is valid.

Answer first

$$(\ln(2+\sqrt{3}/2))/(\sqrt{3}/2)$$

Knowledge points

- continuity of elementary functions
- composition
- continuity of a quotient
- direct substitution

Reading the problem and choosing a method

The expression is composed of sine, cosine, logarithm, square root, and division. Once the logarithm's argument is positive and the denominator is nonzero near the target point, continuity turns the limit into an ordinary function value.

Step-by-step derivation

Step 1. At $x=\pi/6$, $2+\cos x=2+\sqrt{3}/2>0$, so the logarithm is defined and continuous at the corresponding value.

Step 2. Also, $1+\sin(\pi/6)=1+1/2=3/2>0$, so the square root is defined and the denominator $\sqrt{3}/2$ is nonzero.

Step 3. Sine and cosine are continuous on $\mathbb{R}$; adding constants and composing with logarithm and square root preserve continuity here.

Step 4. Because the denominator is nonzero at the target point, the entire quotient is continuous at $x=\pi/6$.

Step 5. Therefore the limit equals $\ln(2+\cos(\pi/6))/\sqrt{1+\sin(\pi/6)}$ by direct substitution.

Step 6. Using $\cos(\pi/6)=\sqrt{3}/2$ and $\sin(\pi/6)=1/2$ gives $\ln(2+\sqrt{3}/2)/\sqrt{3}/2$.

Common pitfalls

- citing elementary continuity without checking the domain and nonzero denominator
- interchanging the values of sine and cosine at $\pi/6$
- making an unjustified algebraic change to the square-root denominator

Verification

At the target point the logarithm's argument is about $2.866 > 0$ and the denominator about $1.225 > 0$, so there is no domain singularity and the substituted value is finite.

Method summary

Direct substitution is justified by continuity; it is not merely a mechanical rule.

Extension prompt

If the logarithm's argument approached 0^+ or the denominator approached 0, continuity would no longer justify direct substitution and a separate limit analysis would be required.

§9 · Operations on Continuous Functions

Standard

Proof

15 min

Textbook-method adaptation

Q053 · The Square Root of a Positive Continuous Function

Question

Suppose f is continuous at a and $f(a) > 0$. Prove that in some neighborhood of a , $f(x) > f(a)/2 > 0$, and then prove that $g(x) = \sqrt{f(x)}$ is continuous at a .

Answer first

Use continuity with $|f(x) - f(a)| < f(a)/2$ to obtain $f(x) > f(a)/2$. Then continuity of the square-root function, or a rationalization estimate, gives $\sqrt{f(x)} \rightarrow \sqrt{f(a)}$.

Knowledge points

- local sign preservation of a continuous function
- epsilon-delta definition
- continuity of the square-root function
- rationalization

Reading the problem and choosing a method

Choose the positive error threshold $f(a)/2$ in the continuity definition to keep $f(x)$ positive nearby. The square root is then defined throughout that neighborhood. Continuity follows either by composition or directly by rationalizing the difference.

Step-by-step derivation

Step 1. Because $f(a) > 0$, the number $f(a)/2$ is a valid positive error threshold.

Step 2. Continuity of f at a gives $\delta_1 > 0$ such that $|x - a| < \delta_1$ implies $|f(x) - f(a)| < f(a)/2$.

Step 3. Therefore $f(x) > f(a) - f(a)/2 = f(a)/2 > 0$. This proves local positivity and ensures that $\sqrt{f(x)}$ is defined.

Step 4. Rationalize: $|g(x) - g(a)| = |\sqrt{f(x)} - \sqrt{f(a)}| = |f(x) - f(a)| / [\sqrt{f(x)} + \sqrt{f(a)}]$.

Step 5. The denominator is at least $\sqrt{f(a)}$, so $|g(x) - g(a)| \leq |f(x) - f(a)| / \sqrt{f(a)}$.

Step 6. Given $\varepsilon > 0$, continuity of f supplies $\delta_2 > 0$ such that $|x - a| < \delta_2$ implies $|f(x) - f(a)| < \varepsilon \sqrt{f(a)}$.

Step 7. Taking $\delta = \min\{\delta_1, \delta_2\}$ simultaneously guarantees a valid square root and $|g(x) - g(a)| < \varepsilon$. Hence g is continuous at a .

Common pitfalls

- writing square roots nearby before proving $f(x) > 0$
- forgetting the denominator in the rationalization step
- using $f(a)/2$ without noting that it is positive

Verification

The same final δ handles both the domain issue and the accuracy requirement, and the last estimate depends only on the controlled quantity $|f(x) - f(a)|$.

Method summary

For a composition with domain restrictions, first use continuity to preserve the needed sign locally, then apply composition or a direct estimate.

Extension prompt

Similarly, if $f(a) \neq 0$, then $f(x)$ stays nonzero in a sufficiently small neighborhood, which proves continuity of $1/f(x)$ at a .

§10 · Continuous Functions on Closed Intervals

Standard

Fill in the blank

6 min

Textbook-method adaptation

Q054 · A Value Between the Endpoint Values

Question

Suppose f is continuous on $[0, 2]$, with $f(0) = -3$ and $f(2) = 5$. By the intermediate value theorem, there is at least one $\xi \in$ such that $f(\xi) = 1$.

Answer first

$(0, 2)$

Knowledge points

- intermediate value theorem
- endpoint values
- attainment in the open interval

Reading the problem and choosing a method

The target value 1 lies strictly between the endpoint values -3 and 5. A continuous function cannot move between those values while skipping 1, and because 1 differs from both endpoint values, the corresponding point is interior.

Step-by-step derivation

Step 1. The function is continuous on the closed interval $[0, 2]$, satisfying the continuity hypothesis of the intermediate value theorem.

Step 2. Numerically, $-3 < 1 < 5$, so $f(0) < 1 < f(2)$.

Step 3. The intermediate value theorem gives some $\xi \in [0, 2]$ with $f(\xi) = 1$.

Step 4. Since $f(0) = -3 \neq 1$ and $f(2) = 5 \neq 1$, this point cannot be either endpoint.

Step 5. Therefore at least one such point satisfies $\xi \in (0, 2)$.

Common pitfalls

- writing only $[0,2]$ without using the fact that the target differs from both endpoint values
- mistaking existence for uniqueness
- failing to verify that the target lies between the endpoint values

Verification

Both endpoints are excluded directly: their values are -3 and 5 , not 1 . Hence the point supplied by the theorem must lie in the interior.

Method summary

The intermediate value theorem guarantees existence, not uniqueness; a target strictly between endpoint values is attained at an interior point.

Extension prompt

For every $\mu \in (-3, 5)$, there is likewise at least one $\xi \in (0, 2)$ such that $f(\xi) = \mu$.

§10 · Continuous Functions on Closed Intervals

Standard

Calculation

14 min

Textbook-method adaptation

Q055 · Locating and Proving Uniqueness of a Cubic Root

Question

Without using derivatives, prove that $x^3+x-1=0$ has exactly one real root in $(1/2, 3/4)$.

Answer first

Let $F(x)=x^3+x-1$. Since $F(1/2)=-3/8<0$ and $F(3/4)=11/64>0$, a root exists in the interval. Since F is strictly increasing on $\mathbb{R}$, that root is unique.

Knowledge points

- zero theorem
- continuity of polynomials
- algebraic proof of strict monotonicity
- separating existence from uniqueness

Reading the problem and choosing a method

The phrase exactly one contains two tasks. Continuity and opposite endpoint signs prove existence. The zero theorem alone does not prove uniqueness, so strict monotonicity must be established separately, here by factoring a difference rather than using derivatives.

Step-by-step derivation

Step 1. Let $F(x)=x^3+x-1$. Being a polynomial, it is continuous on the closed interval $[1/2, 3/4]$.

Step 2. Compute $F(1/2)=1/8+1/2-1=-3/8<0$.

Step 3. Compute $F(3/4)=27/64+3/4-1=27/64+48/64-64/64=11/64>0$.

Step 4. The endpoint values have opposite signs, so the zero theorem gives at least one $\xi \in (1/2, 3/4)$ with $F(\xi)=0$.

Step 5. For uniqueness, let $x>y$. Then $F(x)-F(y)=x^3-y^3+x-y=(x-y)(x^2+xy+y^2+1)$.

Step 6. Here $x-y>0$, and $x^2+xy+y^2=(x+y/2)^2+3y^2/4 \geq 0$, so the second factor is strictly positive.

Step 7. Thus $F(x) > F(y)$, proving that F is strictly increasing on $\mathbb{R}$. A strictly increasing function has at most one zero.

Step 8. Combining at least one with at most one proves that there is exactly one root in $(1/2, 3/4)$.

Common pitfalls

- turning the zero theorem's at least one into exactly one
- making an arithmetic error at the fractional endpoints
- using derivatives when the requested method is pre-derivative

Verification

Continuity and opposite signs are explicitly checked for existence, while the global inequality $F(x) - F(y) > 0$ for arbitrary $x > y$ independently guarantees uniqueness.

Method summary

A standard unique-root proof has two parts: continuity plus a sign change for existence, then strict monotonicity for at most one root.

Extension prompt

The same argument applies to $x^{2m+1} + cx - d = 0$ with $c > 0$: the odd power and positive linear term together give strict increase.

Synthesis

Attempt independently before consulting the solutions.

§1 · Mappings and Functions

Advanced

Proof

10 min

Q056 · Parity of a Composite Function

Question

Let D be symmetric about the origin. Suppose $g:D \rightarrow E$ is even, $f:E \rightarrow R$ is arbitrary, and $h=f \circ g$ is defined on D . Prove that h is even on D .

Answer first

For every $x \in D$, $h(-x) = f(g(-x)) = f(g(x)) = h(x)$; hence h is even.

Knowledge points

- definition of an even function
- composite functions
- symmetric-domain requirement

Reading the problem and choosing a method

The outer function f needs no parity property. Because the even inner function g assigns the same inner value to x and $-x$, f receives the same input in both cases.

Step-by-step derivation

Step 1. Take any $x \in D$. Since D is symmetric about the origin, $-x \in D$ and $h(-x)$ is defined.

Step 2. By the definition of composition, $h(-x) = f(g(-x))$.

Step 3. Since g is even, $g(-x) = g(x)$, so $h(-x) = f(g(x))$.

Step 4. Because $h(x) = f(g(x))$, we obtain $h(-x) = h(x)$ for every $x \in D$; therefore h is even.

Common pitfalls

- omitting the symmetric-domain condition required for parity
- incorrectly assuming that the outer function f must also be even

Verification

For example, with $g(x)=x^2$ and $f(u)=u+u^3$, one gets $h(x)=x^2+x^6$, which indeed satisfies $h(-x)=h(x)$.

Method summary

An even inner function maps x and $-x$ to the same point, so any admissible outer function preserves that symmetry.

Extension prompt

If g is odd, the parity of $f \circ g$ depends on f : an even f gives an even composition, while an odd f gives an odd composition.

§1 · Mappings and Functions

Advanced

Parameter / synthesis / application

14 min

Textbook-method adaptation

Q057 · A Piecewise Parking-Fee Model

Question

A parking lot charges by continuous parking time t in hours. For $0 < t \leq 2$, the fee is 6 yuan. For $2 < t \leq 5$, it charges 6 yuan plus 3 yuan per hour for the portion beyond 2 hours. Write the piecewise formula $C(t)$, find its domain and range, and describe its monotonicity and boundedness.

Answer first

$C(t) = \{6 \text{ for } 0 < t \leq 2; 6 + 3(t-2) = 3t \text{ for } 2 < t \leq 5\}$. The domain is $(0, 5]$, the range is $[6, 15]$, C is nondecreasing, and $6 \leq C(t) \leq 15$.

Knowledge points

- constructing a piecewise model
- range of a piecewise function
- nondecreasing functions
- boundedness

Reading the problem and choosing a method

The pricing rule changes at $t=2$. The first piece is constant, and the second charges only for time beyond 2 hours. The two formulas should also be checked at the junction.

Step-by-step derivation

Step 1. The permitted parking time is $0 < t \leq 5$, so the domain of C is $(0, 5]$.

Step 2. For $0 < t \leq 2$, the fixed charge gives $C(t) = 6$.

Step 3. For $2 < t \leq 5$, the extra time is $t-2$, so $C(t) = 6 + 3(t-2) = 3t$.

Step 4. The first piece contributes the value 6; the second contributes $(6, 15]$, so the total range is $[6, 15]$.

Step 5. C is constant on the first piece and increases on the second, with no decrease across $t=2$; hence it is nondecreasing and satisfies $6 \leq C(t) \leq 15$.

Common pitfalls

- writing $6+3t$ and charging the first two hours twice
- calling the function strictly increasing even though the fee is constant for the first two hours

Verification

At $t=2$ the charge is 6. The right-hand formula tends to $3 \cdot 2 = 6$ near the junction. At $t=5$ it gives 15, matching $6+3 \times 3 = 15$.

Method summary

In modeling, identify the variable range and breakpoints, translate each rule, and then check junctions and global properties.

Extension prompt

If every started half-hour were charged as a full half-hour, the model would become a step function rather than a linear continuous-time model.

§2 · Limits of Sequences

Advanced

Calculation

10 min

Textbook-method adaptation

Q058 · Rationalize a Difference of Radicals

Question

Evaluate $\lim_{n \rightarrow \infty} [\sqrt{n^2+3n}-n]$.

Answer first

 $3/2$.

Knowledge points

- conjugate rationalization
- factoring n^2 from a radical
- limit laws

Reading the problem and choosing a method

The original expression is a difference of two quantities that both grow without bound, so their limits cannot be subtracted. Multiplying by the conjugate removes the cancellation and permits normalization.

Step-by-step derivation

Step 1. Multiply by the conjugate: $\sqrt{n^2+3n}-n=[(n^2+3n)-n^2]/[\sqrt{n^2+3n}+n]$.

Step 2. The numerator simplifies to $3n$, so the expression is $3n/[\sqrt{n^2+3n}+n]$.

Step 3. Since $n>0$, factor n from the radical: $\sqrt{n^2+3n}=n\sqrt{1+3/n}$.

Step 4. Cancel n to obtain $3/[\sqrt{1+3/n}+1]$.

Step 5. As $n \rightarrow \infty$, $3/n \rightarrow 0$, so the denominator tends to 2 and the limit is $3/2$.

Common pitfalls

- subtracting the two infinite limits directly
- factoring n from $\sqrt{n^2+3n}$ without noting that n is positive

Verification

At $n=100$, the original expression is $\sqrt{10300}-100 \approx 1.4889$, close to 1.5; the rationalized form gives the same value.

Method summary

When a difference of radicals contains strong cancellation, use the conjugate to turn it into a manageable quotient.

Extension prompt

Replacing $3n$ by cn gives, by the same argument, $\lim[\sqrt{(n^2+cn)}-n]=c/2$.

§2 · Limits of Sequences

Advanced

Proof

10 min

Textbook-method adaptation

Q059 · Eventual Positivity from a Positive Limit

Question

Suppose $a_n \rightarrow a$ and $a > 0$. Prove that there exists a positive integer N such that $a_n > a/2$ whenever $n > N$.

Answer first

Choose $\varepsilon = a/2$. Since $a_n \rightarrow a$, some N satisfies $|a_n - a| < a/2$ for $n > N$. Hence $a_n > a - a/2 = a/2$.

Knowledge points

- definition of convergence
- eventual sign preservation
- extracting a one-sided bound from an absolute-value inequality

Reading the problem and choosing a method

We want the tail to stay positive by a definite margin. Choosing the allowed error to be half of the positive limit places a_n inside $(a/2, 3a/2)$.

Step-by-step derivation

Step 1. Because $a > 0$, the choice $\varepsilon = a/2$ is valid and positive.

Step 2. From $a_n \rightarrow a$, there exists a positive integer N such that $|a_n - a| < a/2$ whenever $n > N$.

Step 3. This absolute-value inequality is equivalent to $-a/2 < a_n - a < a/2$.

Step 4. Using its left half gives $a_n > a - a/2 = a/2$ for every $n > N$.

Common pitfalls

- claiming eventual positivity without specifying a useful ε
- incorrectly concluding $a_n > a$ from $|a_n - a| < a/2$

Verification

The resulting neighborhood is $(a-a/2, a+a/2) = (a/2, 3a/2)$, whose left endpoint is strictly positive; the conclusion is stronger than mere eventual positivity.

Method summary

To preserve a nonzero sign, choose ϵ as a fixed fraction of the absolute value of the limit.

Extension prompt

If $a < 0$, choosing $\epsilon = |a|/2$ similarly proves that sufficiently large terms satisfy $a - \epsilon < a/2 < 0$.

§2 · Limits of Sequences

Advanced

Multiple choice

8 min

Q060 · Necessary Consequences of a Known Limit

Question

Suppose $a_n \rightarrow 2$. Which conclusions must be true?

Answer first

A, B, D

Knowledge points

- boundedness of convergent sequences
- eventual sign from a positive limit
- limits of subsequences
- convergence does not imply monotonicity

Reading the problem and choosing a method

Convergence keeps the tail close to 2, which ensures boundedness, eventual positivity, and the same limit for every subsequence. However, terms may still oscillate slightly around 2 and need not be monotone.

Step-by-step derivation

Step 1. Every convergent sequence is bounded, so A must be true.

Step 2. Choose $\varepsilon=1$. For all sufficiently large n , $|a_n - 2| < 1$, hence $a_n > 1 > 0$; therefore B is true.

Step 3. The sequence $a_n = 2 + (-1)^n/n$ converges to 2 but alternates above and below 2, so it is not eventually strictly increasing; C need not hold.

Step 4. For any subsequence, its indices still tend to infinity and therefore satisfy the same tail estimates; every subsequence converges to 2, so D is true.

Common pitfalls

- interpreting 'closer and closer' as numerical monotonicity
- remembering boundedness but overlooking that every subsequence inherits the original limit

Verification

In the counterexample $2 + (-1)^n/n$, even terms lie above 2, odd terms below 2, and the error $1/n$ tends to 0, directly disproving the necessity of C.

Method summary

Convergence controls distance to the limit, not the direction or monotone pattern of approach.

Extension prompt

If any subsequence has a limit different from 2, the original sequence cannot converge to 2.

§3 · Limits of Functions

Advanced

Proof

14 min

Textbook-method adaptation

Q061 · An Epsilon-Delta Proof for a Quadratic Limit

Question

Use the epsilon-delta definition to prove $\lim_{x \rightarrow 2} (x^2 + 1) = 5$.

Answer first

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/5\}$. If $0 < |x - 2| < \delta$, then $|x + 2| < 5$, and $|(x^2 + 1) - 5| = |x - 2||x + 2| < 5\delta \leq \varepsilon$.

Knowledge points

- epsilon-delta definition
- use a fixed neighborhood to bound an extra factor
- product error estimate

Reading the problem and choosing a method

The error factors as $|x - 2||x + 2|$. The first factor is directly controlled by δ , while the second varies with x and must first be bounded uniformly in a fixed neighborhood of 2.

Step-by-step derivation

Step 1. Let $\varepsilon > 0$ be arbitrary; the goal is to make $|(x^2 + 1) - 5| < \varepsilon$.

Step 2. Factor the error: $|(x^2 + 1) - 5| = |x^2 - 4| = |x - 2||x + 2|$.

Step 3. First require $|x - 2| < 1$. Then $1 < x < 3$, so $3 < x + 2 < 5$ and hence $|x + 2| < 5$.

Step 4. To satisfy both the fixed-neighborhood bound and the accuracy requirement, choose $\delta = \min\{1, \varepsilon/5\}$.

Step 5. If $0 < |x - 2| < \delta$, then $|(x^2 + 1) - 5| < 5|x - 2| < 5\delta \leq \varepsilon$.

Step 6. Thus an appropriate δ exists for every $\varepsilon > 0$, and the definition yields $\lim_{x \rightarrow 2} (x^2 + 1) = 5$.

Common pitfalls

- setting $\delta = \varepsilon/|x+2|$ and thereby allowing δ to depend on the varying x
- working backward only and failing to verify the chosen δ in the forward direction

Verification

For $\varepsilon=0.1$, choose $\delta=\min\{1,0.02\}=0.02$. If $|x-2|<0.02$, then $|x+2|<4.02<5$ and the error is below $5\times0.02=0.1$.

Method summary

In polynomial epsilon-delta proofs, first use a fixed small neighborhood to bound extra factors, then use δ to meet the requested accuracy.

Extension prompt

A sharper estimate can use $|x+2|<4+\delta$, but the minimum construction is clearer and fully sufficient.

§3 · Limits of Functions

Advanced

Fill in the blank

7 min

Textbook-method adaptation

Q062 · Opposite Signs Near a Vertical Blow-Up

Question

Fill in the blanks: $\lim_{x \rightarrow 2^-} 1/(x-2) = \underline{\hspace{2cm}}$, $\lim_{x \rightarrow 2^+} 1/(x-2) = \underline{\hspace{2cm}}$, and therefore $\lim_{x \rightarrow 2} 1/(x-2) \underline{\hspace{2cm}}$.

Answer first

$-\infty$; $+\infty$; does not exist.

Knowledge points

- infinite limits
- sign of the denominator from each side
- criterion for a two-sided limit

Reading the problem and choosing a method

The denominator tends to zero in magnitude on both sides, but its sign differs: it is a small negative quantity on the left and a small positive quantity on the right. Their reciprocals diverge in opposite directions.

Step-by-step derivation

Step 1. As $x \rightarrow 2^-$, $x-2 < 0$ and $|x-2| \rightarrow 0$, so $1/(x-2)$ is negative with increasing magnitude.

Step 2. Thus $\lim_{x \rightarrow 2^-} 1/(x-2) = -\infty$.

Step 3. As $x \rightarrow 2^+$, $x-2 > 0$ and tends to zero, so its reciprocal becomes arbitrarily large and positive; the right-hand limit is $+\infty$.

Step 4. The one-sided infinite limits have opposite signs and cannot form one two-sided limit, so the two-sided limit does not exist.

Common pitfalls

- writing simply ∞ whenever a denominator tends to zero
- treating $-\infty$ and $+\infty$ as equal because both are unbounded

Verification

At $x=1.99$ the value is -100 , while at $x=2.01$ it is 100 . Closer inputs increase the magnitudes but retain opposite signs.

Method summary

For reciprocal blow-ups, inspect both the direction toward zero and the sign of the denominator.

Extension prompt

For $1/(x-2)^2$ the denominator is positive on both sides, so both one-sided limits are $+\infty$ and the two-sided infinite limit is $+\infty$.

§3 · Limits of Functions

Advanced

Calculation

10 min

Textbook-method adaptation

Q063 · Limit of a Difference of Two Radicals

Question

Evaluate $\lim_{x \rightarrow 0} [\sqrt{1+x} - \sqrt{1-x}]/x$.

Answer first

1.

Knowledge points

- conjugate rationalization
- cancellation in a deleted neighborhood
- limit laws

Reading the problem and choosing a method

Direct substitution gives $0/0$. Because the numerator is a difference of radicals, multiplying by the conjugate sum changes the numerator to $2x$, which can then be canceled.

Step-by-step derivation

Step 1. Multiply by $[\sqrt{1+x} + \sqrt{1-x}]/[\sqrt{1+x} + \sqrt{1-x}]$.

Step 2. The numerator becomes $(1+x) - (1-x) = 2x$.

Step 3. For $x \neq 0$ in the deleted neighborhood, cancel x to obtain $2/[\sqrt{1+x} + \sqrt{1-x}]$.

Step 4. As $x \rightarrow 0$, both radicals tend to 1, so the denominator tends to 2 and the limit is $2/2 = 1$.

Common pitfalls

- splitting the difference into separate limits and remaining stuck at $0/0$
- canceling x without noting that a limit uses $x \neq 0$ in a deleted neighborhood

Verification

At $x=0.01$, the original quotient is approximately $(\sqrt{1.01}-\sqrt{0.99})/0.01 \approx 1.0000$, consistent with 1.

Method summary

When a difference of radicals gives $0/0$, conjugate rationalization often exposes the factor that can be canceled.

Extension prompt

Replacing $1 \pm x$ by $1 \pm ax$ gives, by the same method, the limit a .

§4 · Infinitesimals and Infinite Quantities

Advanced

Calculation

14 min

Q064 · Parameter Classification with an Oscillatory Factor

Question

Let $f_p(x) = |x|^p \sin(1/x)$ for $x \neq 0$. For $p > 0$, $p = 0$, and $p < 0$, decide whether f_p is infinitesimal, bounded with no limit, or unbounded without tending to $+\infty$ or $-\infty$ as $x \rightarrow 0$.

Answer first

For $p > 0$, $f_p(x) \rightarrow 0$ and is infinitesimal. For $p = 0$, it is bounded but has no limit. For $p < 0$, it is unbounded and tends to neither $+\infty$ nor $-\infty$.

Knowledge points

- Bounded oscillatory factors
- Squeeze theorem
- Construction of approaching sequences
- Unboundedness versus an infinite limit

Reading the problem and choosing a method

The key bound is $|\sin(1/x)| \leq 1$. For $p > 0$ the power suppresses oscillation; for $p \leq 0$, choose sequences on which the sine equals 1, -1 , or 0.

Step-by-step derivation

Step 1. If $p > 0$, then $|f_p(x)| \leq |x|^p$ and $|x|^p \rightarrow 0$; the squeeze theorem gives $f_p(x) \rightarrow 0$.

Step 2. If $p = 0$, then $f_0(x) = \sin(1/x)$, which always lies in $[-1, 1]$, so it is bounded.

Step 3. Let $x_n = 1/(\pi/2 + 2\pi n)$ and $y_n = 1/(3\pi/2 + 2\pi n)$. Both tend to 0, but $f_0(x_n) = 1$ and $f_0(y_n) = -1$, so no limit exists.

Step 4. If $p < 0$, then $f_p(x_n) = |x_n|^p \rightarrow +\infty$ while $f_p(y_n) = -|y_n|^p \rightarrow -\infty$, so the function is unbounded.

Step 5. Also let $z_n = 1/(\pi n)$; then $f_p(z_n) = 0$. Hence the whole function tends to neither $+\infty$ nor $-\infty$.

Common pitfalls

- Ignoring the zeros and sign changes of $\sin(1/x)$
- Equating unboundedness with an infinite limit
- Using one sequence as if it established a full function limit

Verification

All three sequences approach 0 while forcing the sine to be 1, -1 , and 0. This confirms oscillation for $p \leq 0$, whereas the absolute bound tends directly to 0 for $p > 0$.

Method summary

A bounded factor times an infinitesimal tends to 0; an unbounded factor times an oscillatory term requires sequence-by-sequence analysis.

Extension prompt

Replacing $|x|^p$ by a positive $a(x)$: if $a(x) \rightarrow 0$, the product tends to 0; if $a(x) \rightarrow \infty$, sequences usually decide whether the result is merely unbounded rather than infinite.

§4 · Infinitesimals and Infinite Quantities

Advanced

True/false with justification

10 min

Q065 · Does Local Unboundedness Imply an Infinite Limit?

Question

True or false, with justification: If f is unbounded in every deleted neighborhood of $x=0$, then $\lim_{x \rightarrow 0} |f(x)| = +\infty$.

Answer first

False. Let $f(x)=1/|x|$ for rational x and $f(x)=0$ for irrational x . It is unbounded in every deleted neighborhood, but remains 0 along irrational points approaching 0.

Knowledge points

- Local unboundedness
- Definition of divergence to infinity
- Dense-set counterexamples
- Necessary versus sufficient conditions

Reading the problem and choosing a method

Local unboundedness only asks for some arbitrarily large values in every neighborhood. A limit of $+\infty$ requires every sufficiently nearby value to exceed each threshold, which is much stronger.

Step-by-step derivation

Step 1. Define $f(x)=1/|x|$ for rational x and $f(x)=0$ for irrational x , with $x \neq 0$.

Step 2. Given $\delta > 0$ and $M > 0$, choose a rational x with $0 < |x| < \min(\delta, 1/M)$; then $f(x)=1/|x| > M$, so f is unbounded in every deleted neighborhood.

Step 3. If $|f(x)| \rightarrow +\infty$, then for $M=1$ there must be $\delta > 0$ such that every x with $0 < |x| < \delta$ satisfies $|f(x)| > 1$.

Step 4. Every such neighborhood contains an irrational y , but $f(y)=0 \leq 1$, so the universal condition fails.

Step 5. Hence local unboundedness does not imply an infinite absolute-value limit, and the statement is false.

Common pitfalls

- Confusing the existence of a large value with all values being large
- Looking only along rational points
- Giving a counterexample without proving unboundedness in every neighborhood

Verification

Along rational $x_n=1/n$, $f(x_n)=n \rightarrow \infty$; along any irrational $y_n \rightarrow 0$, $f(y_n)=0$. These two behaviors rule out $|f(x)| \rightarrow \infty$.

Method summary

Unboundedness is existential, while an infinite limit is an eventual universal statement.

Extension prompt

If $|f(x)| \rightarrow \infty$, then f is locally unbounded; the converse is generally false.

§5 · Algebra of Limits

Advanced

Calculation

14 min

Q066 · Double Rationalization of Nested Radicals

Question

Evaluate $\lim_{x \rightarrow 0} [\sqrt{(1+\sqrt{(1+x))}-\sqrt{2}}]/x$ without derivatives or Taylor expansions.

Answer first

$$1/(4\sqrt{2})$$

Knowledge points

- Two successive conjugate rationalizations
- 0/0 limits
- Product and quotient limit laws

Reading the problem and choosing a method

Rationalizing the outer radical still leaves $\sqrt{(1+x)-1}$, so a second rationalization is needed to expose the factor x .

Step-by-step derivation

Step 1. Call the limit L and multiply first by the outer conjugate $\sqrt{(1+\sqrt{(1+x))}+\sqrt{2}}$.

Step 2. The numerator becomes $\sqrt{(1+x)-1}$, so L equals $[\sqrt{(1+x)-1}]/\{x[\sqrt{(1+\sqrt{(1+x))}+\sqrt{2})]\}$.

Step 3. Rationalize $\sqrt{(1+x)-1}$ with $\sqrt{(1+x)+1}$, obtaining $x/[\sqrt{(1+x)+1}]$.

Step 4. Cancel x to get $L=1/[\{\sqrt{(1+x)+1}\}[\sqrt{(1+\sqrt{(1+x))}+\sqrt{2})]]$.

Step 5. As $x \rightarrow 0$, the brackets tend to 2 and $2\sqrt{2}$, so $L=1/(4\sqrt{2})$.

Common pitfalls

- Stopping after only one rationalization
- Losing the term $\sqrt{(1+x)-1}$
- Multiplying $2 \cdot 2\sqrt{2}$ incorrectly

Verification

At $x=0.001$ the original expression is about 0.17672, while $1/(4\sqrt{2})\approx 0.17678$.

Method summary

Nested radicals often require layer-by-layer rationalization aimed at exposing a linear factor.

Extension prompt

The same strategy can be repeated for deeper nests, while tracking the limit of each conjugate factor.

§5 · Algebra of Limits

Advanced

Calculation

14 min

Q067 · Determining Parameters from a Finite Limit

Question

Find real numbers a, b such that $\lim_{x \rightarrow 1} (x^2 + ax + b)/(x - 1) = 3$.

Answer first

$a = 1$ and $b = -2$.

Knowledge points

- Necessary numerator cancellation for a finite limit
- Polynomial factorization
- Parameter equations

Reading the problem and choosing a method

The denominator tends to 0. A finite quotient limit forces the numerator to tend to 0 first; then match the limit after cancellation.

Step-by-step derivation

Step 1. A finite limit requires the numerator at $x=1$ to vanish, so $1+a+b=0$.

Step 2. Using $b=-1-a$, rewrite the numerator as $x^2+ax-1-a$.

Step 3. Factor $x^2+ax-1-a=(x-1)(x+a+1)$.

Step 4. For $x \neq 1$ the quotient is $x+a+1$, whose limit at 1 is $a+2$.

Step 5. Set $a+2=3$ to get $a=1$, then $1+a+b=0$ gives $b=-2$.

Common pitfalls

- Stopping after $1+a+b=0$
- Treating the necessary cancellation as sufficient by itself
- Making a sign error in factorization

Verification

With $a=1, b=-2$, the numerator is $x^2+x-2=(x-1)(x+2)$; the quotient tends to 3.

Method summary

Parameter-dependent $0/0$ limits usually require two stages: cancel the constant mismatch, then match the reduced limit.

Extension prompt

If the target is L , the same work gives $a=L-2$ and $b=1-L$.

§5 · Algebra of Limits

Advanced

Calculation

12 min

Q068 · Using a Known Ratio of Deviations

Question

As $x \rightarrow a$, suppose $f(x) \rightarrow 1$, $g(x) \rightarrow 2$, and $[f(x)-1]/[g(x)-2] \rightarrow 3$. Find $\lim_{x \rightarrow a} [f(x)^2-1]/[g(x)^2-4]$.

Answer first

 $3/2$

Knowledge points

- Difference-of-squares factorization
- Product limit law
- Preserving a known ratio

Reading the problem and choosing a method

Do not take the two squared differences separately, because that remains $0/0$. Factor them and preserve the given ratio as one factor.

Step-by-step derivation

Step 1. Factor $f^2-1=(f-1)(f+1)$ and $g^2-4=(g-2)(g+2)$.

Step 2. In the relevant deleted neighborhood, write the quotient as $[(f-1)/(g-2)] \cdot [(f+1)/(g+2)]$.

Step 3. The first factor tends to 3 by hypothesis.

Step 4. The second tends to $(1+1)/(2+2)=2/4=1/2$, with denominator limit $4 \neq 0$.

Step 5. The product law yields $3 \cdot 1/2 = 3/2$.

Common pitfalls

- Calling the $0/0$ squared-difference quotient zero
- Dropping $f+1$ or $g+2$
- Inverting the given ratio

Verification

Set $g-2=t$ and $f-1=3t$. Then the quotient is $(6t+9t^2)/(4t+t^2) \rightarrow 6/4=3/2$.

Method summary

When a ratio of deviations is given, factor expressions so that the known ratio remains intact.

Extension prompt

The same idea handles $[f^n-1]/[g^m-2^m]$ using difference-of-powers identities.

§5 · Algebra of Limits

Advanced

Proof

20 min

Textbook-method
adaptation

Q069 · Proving the Product Limit Theorem from Definitions

Question

Suppose $f(x) \rightarrow A$ and $g(x) \rightarrow B$ as $x \rightarrow a$. Using only the limit definition and local boundedness of convergent functions, prove that $f(x)g(x) \rightarrow AB$.

Answer first

Use $f(x)g(x) - AB = f(x)[g(x) - B] + B[f(x) - A]$ and control the two terms separately.

Knowledge points

- Product limit theorem
- Local boundedness of convergent functions
- Error decomposition
- Epsilon-delta proof

Reading the problem and choosing a method

The decomposition $f(g-B) + B(f-A)$ needs only a uniform bound on f , followed by a split of the epsilon budget.

Step-by-step derivation

Step 1. Since $f(x) \rightarrow A$, choose a deleted neighborhood where $|f(x) - A| < 1$; then $|f(x)| \leq |A| + 1$. Set $M = |A| + 1 > 0$.

Step 2. Given $\varepsilon > 0$, $g(x) \rightarrow B$ gives a neighborhood where $|g(x) - B| < \varepsilon / (2M)$.

Step 3. Also $f(x) \rightarrow A$ gives a neighborhood where $|f(x) - A| < \varepsilon / [2(|B| + 1)]$.

Step 4. Take the minimum of the three neighborhood radii so that all estimates hold simultaneously.

Step 5. Using $fg - AB = f(g - B) + B(f - A)$, obtain $|fg - AB| \leq M \cdot \varepsilon / (2M) + |B| \cdot \varepsilon / [2(|B| + 1)] < \varepsilon$.

Step 6. Therefore $\lim_{x \rightarrow a} f(x)g(x) = AB$ by definition.

Common pitfalls

- Assuming boundedness without deriving it
- Dividing by $|B|$ when $B=0$
- Failing to take the minimum of several deltas
- Dropping a term in the error decomposition

Verification

Each error contribution is below $\varepsilon/2$, and the denominators M and $|B|+1$ are always positive.

Method summary

Definition proofs of limit laws often first establish a local bound and then split the total error into controllable pieces.

Extension prompt

Repeated application proves that the limit of any finite product equals the product of the limits.

§6 · Existence
Criteria and Two
Fundamental Limits

Advanced

Calculation

8 min

Textbook-method
adaptation

Q070 · A Ratio of Sines with Different Scales

Question

Evaluate $\lim_{x \rightarrow 0} \sin(3x)/\sin(5x)$.

Answer first

 $3/5$

Knowledge points

- First fundamental limit
- Multiplicative decomposition
- Reciprocal of a nonzero limit

Reading the problem and choosing a method

Create $\sin u/u$ separately in numerator and denominator, retaining the scale ratio $3/5$.

Step-by-step derivation

Step 1. Write $\sin(3x)/\sin(5x)=[\sin(3x)/(3x)] \cdot (3/5) \cdot [5x/\sin(5x)]$.Step 2. As $x \rightarrow 0$, both $3x$ and $5x$ tend to 0.

Step 3. The first bracket tends to 1.

Step 4. Since $\sin(5x)/(5x) \rightarrow 1$, its reciprocal $5x/\sin(5x)$ also tends to 1.Step 5. Therefore the limit is $1 \cdot (3/5) \cdot 1 = 3/5$.

Common pitfalls

- Replacing $\sin(3x)/\sin(5x)$ by $\sin(3/5)$
- Dropping the scale ratio
- Taking a reciprocal without a nonzero limit

Verification

At $x=0.001$, $\sin 0.003/\sin 0.005 \approx 0.6000016$, close to 0.6.

Method summary

The limiting ratio of small-angle sines equals the ratio of their linear scales.

Extension prompt

For $a, b \neq 0$, $\lim_{x \rightarrow 0} \sin(ax)/\sin(bx) = a/b$.

§6 · Existence
Criteria and Two
Fundamental Limits

Advanced

Calculation

12 min

Textbook-method
adaptation

Q071 · The Second Fundamental Limit with a Fractional Base

Question

Evaluate $\lim_{x \rightarrow 0} [(1+x)/(1-x)]^{1/x}$.

Answer first

e^2

Knowledge points

- The 1^∞ form
- Second fundamental limit
- Rewriting the increment

Reading the problem and choosing a method

Write the fractional base as 1 plus $t=2x/(1-x)$, then compute the limit of t times the original exponent $1/x$.

Step-by-step derivation

Step 1. $(1+x)/(1-x)=1+2x/(1-x)$. Set $t=2x/(1-x)$, so $t \rightarrow 0$.

Step 2. Rewrite the expression as $(1+t)^{1/x}=[(1+t)^{1/t}]^{t/x}$.

Step 3. The second fundamental limit gives $(1+t)^{1/t} \rightarrow e$.

Step 4. Also $t/x=2/(1-x) \rightarrow 2$.

Step 5. Therefore the full limit is e^2 .

Common pitfalls

- Replacing the increment by $2x$ exactly
- Concluding 1 because the base tends to 1
- Dropping $1-x$ when computing t/x

Verification

At $x=0.001$, $[(1.001)/(0.999)]^{1000} \approx 7.38906$, close to e^2 .

Method summary

For a complicated 1^∞ form, write the base as $1+t$ and find the limit of t times the original exponent.

Extension prompt

The same method gives $\lim_{(x \rightarrow 0)} [(1+ax)/(1+bx)]^{1/x} = e^{a-b}$.

§6 · Existence Criteria and Two Fundamental Limits

Advanced

Calculation

12 min

Q072 · Determining Parameters from an Exponential Limit

Question

Let a, b be positive real numbers. Given $\lim_{x \rightarrow 0} (1+ax)^{b/x} = e^6$ and $a+b=5$, find all ordered pairs (a, b) .

Answer first

$(a, b) = (2, 3)$ or $(3, 2)$.

Knowledge points

- Parameterized second fundamental limit
- Injectivity of the exponential
- Quadratic equation

Reading the problem and choosing a method

Standardization gives the limit e^{ab} . Comparing with e^6 yields $ab=6$, which combines with $a+b=5$.

Step-by-step derivation

Step 1. Set $u=ax$; then $(1+ax)^{b/x} = [(1+u)^{1/u}]^{ab}$.

Step 2. As $x \rightarrow 0$, $u \rightarrow 0$, so the limit is e^{ab} .

Step 3. Since $e^{ab} = e^6$ and the exponential is one-to-one, $ab=6$.

Step 4. Together with $a+b=5$, a and b are the two positive roots of $t^2-5t+6=0$.

Step 5. Factoring gives $(t-2)(t-3)=0$, hence $(a, b) = (2, 3)$ or $(3, 2)$.

Common pitfalls

- Using b/a instead of ab
- Giving only one ordered pair
- Ignoring positivity and local positivity of the base

Verification

Both pairs have product 6 and sum 5, and substitution into the general limit gives e^6 .

Method summary

For a parameterized 1^∞ limit, extract the increment-exponent product and convert the result into algebraic conditions.

Extension prompt

Without the sum condition, every positive pair satisfying $ab=6$ would work.

§6 · Existence
Criteria and Two
Fundamental Limits

Advanced

Proof

14 min

Textbook-method
adaptation

Q073 · Squeezing the Floor-Function Error

Question

Use the squeeze theorem to prove $\lim_{x \rightarrow +\infty} \lfloor x \rfloor / x = 1$.

Answer first

From $x-1 < \lfloor x \rfloor \leq x$ and $x > 0$, obtain $1-1/x < \lfloor x \rfloor / x \leq 1$; both bounds tend to 1.

Knowledge points

- Basic floor-function inequalities
- Squeeze at infinity
- Division by a positive quantity

Reading the problem and choosing a method

The error between $\lfloor x \rfloor$ and x is less than 1. Dividing this bounded absolute error by the growing scale x makes the relative error vanish.

Step-by-step derivation

Step 1. By definition, $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ for every real x .

Step 2. The right inequality gives $x-1 < \lfloor x \rfloor$.

Step 3. For $x \rightarrow +\infty$ we eventually have $x > 0$; divide $x-1 < \lfloor x \rfloor \leq x$ by x to get $1-1/x < \lfloor x \rfloor / x \leq 1$.

Step 4. The lower and upper bounds both tend to 1.

Step 5. The squeeze theorem yields $\lfloor x \rfloor / x \rightarrow 1$.

Common pitfalls

- Writing $\lfloor x \rfloor = x - 1$ identically
- Dividing before confirming $x > 0$
- Mentioning a bounded error without converting it to a relative error

Verification

The proof gives $0 \leq 1 - \lfloor x \rfloor / x < 1/x$, whose right side explicitly tends to 0.

Method summary

A uniformly bounded absolute error becomes a vanishing relative error after division by a scale tending to infinity.

Extension prompt

Similarly, $\lim_{n \rightarrow \infty} \lfloor an \rfloor / n = a$ for every $a > 0$.

§6 · Existence
Criteria and Two
Fundamental Limits

Advanced

Proof

18 min

Textbook-method
adaptation

Q074 · A Monotone-Bounded Proof for a Linear Recurrence

Question

Let $a_1 > 2$ and $a_{n+1} = (a_n + 2)/3$. Prove that $\{a_n\}$ converges and find its limit, without first deriving a closed form.

Answer first

$\{a_n\}$ is decreasing and bounded below by 1, hence convergent; its limit is 1.

Knowledge points

- Mathematical induction
- Monotone convergence theorem
- Limit equation for a recurrence

Reading the problem and choosing a method

First prove $a_n > 1$ to obtain a lower bound and determine the sign of $a_{n+1} - a_n$. Then apply monotone convergence and only afterward pass to the limit.

Step-by-step derivation

Step 1. Since $a_1 > 2 > 1$, and $a_n > 1$ implies $a_{n+1} = (a_n + 2)/3 > (1 + 2)/3 = 1$, induction gives $a_n > 1$ for all n .

Step 2. Compute $a_{n+1} - a_n = (a_n + 2 - 3a_n)/3 = 2(1 - a_n)/3 < 0$.

Step 3. Thus $\{a_n\}$ is decreasing and bounded below by 1.

Step 4. The monotone convergence theorem guarantees convergence. Let the limit be L .

Step 5. Passing to the limit in the recurrence gives $L = (L + 2)/3$, so $3L = L + 2$ and $L = 1$.

Step 6. Therefore the sequence converges to 1.

Common pitfalls

- Declaring the difference negative before proving $a_n > 1$
- Solving a limit equation before proving existence
- Calling the lower bound 1 an attained minimum

Verification

For $a_1=4$, the first terms are $4, 2, 4/3, 10/9$; they remain above 1 and decrease toward 1.

Method summary

For a recursive sequence: preserve a bound, determine monotonicity, prove convergence, then solve the limit equation.

Extension prompt

Indeed, $a_{n+1} - 1 = (a_n - 1)/3$, so the error shrinks by a factor of $1/3$.

§7 · Comparison of
Infinitesimals

Advanced

Calculation

11 min

Q075 · A Mixed Exponential-Logarithmic-Trigonometric Limit

Question

Evaluate $\lim_{x \rightarrow 0} [(e^{2x} - 1)\tan(3x)]/[x \ln(1 + 4x)]$.

Answer first

 $3/2$

Knowledge points

- $e^u - 1 \sim u$
- $\tan u \sim u$
- $\ln(1 + u) \sim u$
- Tracking constants

Reading the problem and choosing a method

All four vanishing factors occur in a product-quotient structure, so factorwise substitution is valid. The challenge is preserving the scales 2, 3, and 4.

Step-by-step derivation

Step 1. $e^{2x} - 1 \sim 2x$.Step 2. $\tan(3x) \sim 3x$.Step 3. $\ln(1 + 4x) \sim 4x$.Step 4. Thus the expression is equivalent to $[(2x)(3x)]/[x(4x)]$.Step 5. Cancel x^2 to obtain $6/4 = 3/2$.

Common pitfalls

- Replacing every composite function by x
- Dropping the external x in the denominator
- Canceling an internal x before forming valid ratios

Verification

Factor the expression into three standard ratios and the constant $3/2$; every standard ratio tends to 1.

Method summary

For mixed equivalences, pair factors one by one and keep a clear ledger of scale constants.

Extension prompt

Replacing 2,3,4 by a,b,c , with $c \neq 0$, gives the limit ab/c .

§7 · Comparison of Infinitesimals

Advanced

Calculation

12 min

Q076 · Determining Parameters from an Equivalence

Question

Let $a, b > 0$. If $e^{ax} - 1 \sim b \sin(2x)$ as $x \rightarrow 0$ and $a + b = 3$, find a and b .

Answer first

$a = 2$ and $b = 1$.

Knowledge points

- Ratio definition of equivalence
- Composite scaling
- Parameter equations

Reading the problem and choosing a method

Reduce both sides to their first-order principal quantities: $e^{ax} - 1 \sim ax$ and $b \sin 2x \sim 2bx$. Equivalence requires equal leading coefficients.

Step-by-step derivation

Step 1. From $e^u - 1 \sim u$, obtain $e^{ax} - 1 \sim ax$.

Step 2. From $\sin u \sim u$, obtain $b \sin(2x) \sim 2bx$.

Step 3. Their ratio must tend to 1, so $a/(2b) = 1$ and $a = 2b$.

Step 4. Combine this with $a + b = 3$ to get $2b + b = 3$.

Step 5. Hence $b = 1$ and $a = 2$.

Common pitfalls

- Using b instead of $2b$ for $b \sin 2x$
- Stopping at $a=2b$
- Confusing equivalence with merely having a finite nonzero ratio

Verification

For $a=2, b=1$, $(e^{(2x)}-1)/\sin 2x = [(e^{(2x)}-1)/(2x)] \cdot [2x/\sin 2x] \rightarrow 1$.

Method summary

Parameterized equivalence problems reduce to matching the coefficients of the lowest-order nonzero principal quantities.

Extension prompt

For merely the same order, $a/(2b)$ need only be finite and nonzero, not necessarily 1.

§7 · Comparison of Infinitesimals

Advanced

Calculation

12 min

Q077 · Checking a First-Order Remainder in a Difference

Question

Without Taylor expansions, evaluate $\lim_{x \rightarrow 0} [\sqrt{1+x} - 1 - x/2]/x$.

Answer first

0

Knowledge points

- Rationalization
- Care with difference structures
- Remainder after an infinitesimal equivalence

Reading the problem and choosing a method

One should not merely replace $\sqrt{1+x} - 1$ by $x/2$ inside a difference. Instead, simplify the full quotient exactly.

Step-by-step derivation

Step 1. Split the expression as $[\sqrt{1+x} - 1]/x - 1/2$.

Step 2. Rationalize the first term: $[\sqrt{1+x} - 1]/x = 1/[\sqrt{1+x} + 1]$ for $x \neq 0$.

Step 3. Thus the expression is $1/[\sqrt{1+x} + 1] - 1/2$.

Step 4. As $x \rightarrow 0$, the first term tends to $1/(1+1) = 1/2$.

Step 5. The requested limit is $1/2 - 1/2 = 0$.

Common pitfalls

- Substituting inside a difference without justification
- Dropping +1 after rationalization
- Misreading the denominator as x^2

Verification

Exact combination gives $[1-\sqrt{(1+x)}]/[2(\sqrt{(1+x)}+1)]$, whose numerator tends to 0 and denominator to 4.

Method summary

An equivalence may suggest the answer, but differences should be handled by exact identities or regrouping into known ratios.

Extension prompt

With x^2 in the denominator the limit is not 0; higher-order information is needed, as shown in the comprehensive problem.

§8 · Continuity and Discontinuities

Advanced

Calculation

16 min

Textbook-method adaptation

Q078 · Three Types of Discontinuity in One Function

Question

Let $f(x) = -1$ for $x < 0$; $f(x) = (x^2 - 1)/(x - 1)$ for $0 \leq x < 2$ with $x \neq 1$; and $f(x) = 1/(x - 2)$ for $x > 2$. Find and classify every discontinuity. If one is removable, give the value for a continuous extension.

Answer first

$x=0$ is a jump discontinuity; $x=1$ is removable with continuous-extension value 2; $x=2$ is an infinite discontinuity of the second kind.

Knowledge points

- junctions of a piecewise function
- canceling a common factor
- one-sided limits
- first- and second-kind discontinuities

Reading the problem and choosing a method

The only candidates are the formula junctions 0, 2 and the zero 1 of the denominator. Every other point lies in the interior of a continuous piece. We classify each candidate from its two one-sided limits.

Step-by-step derivation

Step 1. For $0 \leq x < 2$ and $x \neq 1$, $(x^2 - 1)/(x - 1) = x + 1$. Thus each piece is continuous away from 0, 1, 2.

Step 2. At $x=0$, $\lim_{x \rightarrow 0^-} f(x) = -1$ while $\lim_{x \rightarrow 0^+} f(x) = 1$. The finite one-sided limits are unequal, so 0 is a jump discontinuity.

Step 3. At $x=1$, both sides use the simplified expression $x+1$, giving $\lim_{x \rightarrow 1} f(x) = 2$. Since $f(1)$ is undefined, 1 is removable.

Step 4. The unique continuous extension at $x=1$ is obtained by defining $f(1) = 2$.

Step 5. At $x=2$, $\lim_{x \rightarrow 2^-} f(x) = 3$, whereas $\lim_{x \rightarrow 2^+} f(x) = +\infty$.

Step 6. Because at least one one-sided limit is not finite, $x=2$ is an infinite discontinuity of the second kind.

Step 7. Therefore the complete list is 0,1,2, with the classifications stated in the answer.

Common pitfalls

- assuming cancellation makes the original function defined at $x=1$
- looking only at point values instead of one-sided limits
- misclassifying $x=2$ as a jump because its left-hand limit is finite

Verification

Every point outside $\{0,1,2\}$ is internal to one continuous formula. The one-sided limits at the three candidates exhibit exactly the jump, removable, and infinite patterns, so the list is complete.

Method summary

First locate junctions, undefined points, and dangerous points of elementary functions; then classify them with one-sided limits.

Extension prompt

If the third piece is replaced by $1/(x-2)^2$, the right-hand limit as $x \rightarrow 2^+$ is still $+\infty$, while the left-hand limit from the middle piece remains 3; thus $x=2$ is still an infinite discontinuity of the second kind.

§8 · Continuity and Discontinuities

Advanced

Error diagnosis

14 min

Q079 · Must the Sum of Two Discontinuous Functions Be Discontinuous?

Question

A student claims: “If both f and g are discontinuous at $x=0$, then $f+g$ must be discontinuous at 0, because only sums of continuous functions are continuous.” Identify the logical error and construct and verify a counterexample.

Answer first

The claim is false; the theorem “ f, g continuous implies $f+g$ continuous” cannot be reversed. Let $f(0)=0$ and $f(x)=1$ for $x \neq 0$; let $g(0)=0$ and $g(x)=-1$ for $x \neq 0$. Both are discontinuous at 0, but $f+g \equiv 0$ is continuous.

Knowledge points

- sufficient conditions and converses
- algebra of continuous functions
- counterexample construction
- single-point removable discontinuity

Reading the problem and choosing a method

The algebra theorem is a forward sufficient condition: if both summands are continuous, their sum is continuous. It does not say that discontinuity of the summands must survive addition, because their defects can cancel.

Step-by-step derivation

Step 1. The valid implication is “ f, g continuous at 0 $\Rightarrow f+g$ continuous at 0.” The student has used an invalid converse rather than the actual contrapositive.

Step 2. Define $f(0)=0$ and $f(x)=1$ for $x \neq 0$. Then $\lim_{x \rightarrow 0} f(x)=1 \neq f(0)=0$, so f is discontinuous at 0.

Step 3. Define $g(0)=0$ and $g(x)=-1$ for $x \neq 0$. Then $\lim_{x \rightarrow 0} g(x)=-1 \neq g(0)=0$, so g is also discontinuous at 0.

Step 4. For $x \neq 0$, $(f+g)(x)=1+(-1)=0$; at $x=0$, $(f+g)(0)=0+0=0$.

Step 5. Thus $(f+g)(x)=0$ for every real x , so the sum is a constant function and is continuous at 0.

Step 6. This counterexample disproves the claim and exhibits exact cancellation of two removable defects.

Common pitfalls

- treating a sufficient condition as a necessary one
- failing to verify that both component functions are actually discontinuous
- checking the sum only for $x \neq 0$ and forgetting its value at 0

Verification

Direct checks give limit 1 but value 0 for f , limit -1 but value 0 for g , and both limit and value 0 for the sum. All three conclusions follow from the definition.

Method summary

Continuity algebra works forward; it does not guarantee that discontinuity is preserved by algebraic operations.

Extension prompt

Even jump behavior may partly cancel. Always analyze the limit and point value of the resulting function rather than relying only on labels attached to the summands.

§8 · Continuity and Discontinuities

Advanced

Proof

20 min

Textbook-method adaptation

Q080 · Finding the Unique Junction Value by One-Sided Continuity

Question

Fix a real number a . Define $f(x)=\sin(x-a)/(x-a)$ for $x<a$, $f(a)=c$, and $f(x)=\ln(1+x-a)/(x-a)$ for $x>a$. First prove that a piecewise function is continuous at a exactly when both one-sided limits equal $f(a)$, and then find the unique c making this function continuous.

Answer first

$c=1$

Knowledge points

- equivalence of continuity and one-sided continuity
- standard limits
- joining a piecewise function
- uniqueness of continuous extension

Reading the problem and choosing a method

The proof uses the equivalence between a two-sided limit and two equal one-sided limits. For the application, set $t=x-a$ on each side and use the two standard limits. Their common finite value is the only possible junction value.

Step-by-step derivation

Step 1. In general, if f is continuous at a , then $\lim_{x \rightarrow a} f(x)=f(a)$. Existence of this two-sided limit implies that both one-sided limits exist and equal $f(a)$.

Step 2. Conversely, if $\lim_{x \rightarrow a^-} f(x)=f(a)$ and $\lim_{x \rightarrow a^+} f(x)=f(a)$, the two one-sided limits exist and agree. Hence the two-sided limit exists and equals $f(a)$, so f is continuous at a .

Step 3. For the left piece, put $t=x-a$. As $x \rightarrow a^-$, $t \rightarrow 0^-$, and $\lim_{x \rightarrow a^-} \sin(x-a)/(x-a)=\lim_{t \rightarrow 0^-} \sin t/t=1$.

Step 4. For the right piece, again put $t=x-a$. As $x \rightarrow a^+$, $t \rightarrow 0^+$, and $\lim_{(x \rightarrow a^+)} \ln(1+x-a)/(x-a) = \lim_{(t \rightarrow 0^+)} \ln(1+t)/t = 1$.

Step 5. Both one-sided limits are 1. By the criterion just proved, continuity requires and is guaranteed by $f(a)=c=1$.

Step 6. The value is unique: if $c \neq 1$, the two-sided limit remains 1 but no longer equals the point value.

Common pitfalls

- checking equality of the one-sided limits but not comparing them with c
- reversing a one-sided direction after substitution
- incorrectly assigning limit 0 to $\ln(1+t)/t$

Verification

With $c=1$, the left-hand limit, right-hand limit, and function value are all 1. Any other c violates the necessary equality between limit and value.

Method summary

At a piecewise junction, compute both one-sided limits and compare their common value with the actual point value.

Extension prompt

If the right piece is replaced by $(e^{(x-a)}-1)/(x-a)$, its right-hand limit is still 1, so the unique junction value is unchanged.

§10 · Continuous Functions on Closed Intervals

Advanced

Calculation

16 min

Textbook-method adaptation

Q081 · Finding Closed-Interval Extrema without Derivatives

Question

Without using derivatives, find the maximum and minimum of $f(x)=x^4-2x^2+3$ on $[-2,2]$, together with all points where they are attained.

Answer first

The minimum is 2, attained at $x=\pm 1$; the maximum is 11, attained at $x=\pm 2$.

Knowledge points

- extreme value theorem
- completing the square
- range under substitution
- equality conditions

Reading the problem and choosing a method

Continuity guarantees that extrema are attained, but it does not compute them. Put $t=x^2$ to turn the quartic into a quadratic on $[0,4]$, then use a completed square to obtain both bounds and their equality cases.

Step-by-step derivation

Step 1. As a polynomial, f is continuous on the closed interval $[-2,2]$, so the extreme value theorem guarantees a maximum and a minimum.

Step 2. Set $t=x^2$. Since $x\in[-2,2]$, the new variable satisfies $t\in[0,4]$.

Step 3. Rewrite $f=t^2-2t+3=(t-1)^2+2$, which gives $f\geq 2$.

Step 4. Equality holds exactly when $t=1$, or $x^2=1$. Hence the minimum 2 is attained at $x=\pm 1$.

Step 5. For the upper bound, $t\in[0,4]$ implies $t-1\in[-1,3]$, and therefore $(t-1)^2\leq 9$.

Step 6. The square reaches 9 exactly at $t=4$, so $f\leq 11$, with $x^2=4$ and therefore $x=\pm 2$.

Step 7. Thus the minimum is 2 and the maximum is 11, at the stated points.

Common pitfalls

- writing the range of $t=x^2$ as $[-4,4]$
- finding only the minimum from the completed square and forgetting the maximum
- failing to check whether equality is attainable in the original interval

Verification

Direct substitution gives $f(\pm 1)=1-2+3=2$ and $f(\pm 2)=16-8+3=11$. The substitution interval includes every possible value of x^2 , so no interior point is omitted.

Method summary

The extreme value theorem guarantees attainment; algebraic transformation computes the values and equality locations.

Extension prompt

Even polynomials of this type can often be reduced by setting $t=x^2$, but the exact range of t must be written first.

§10 · Continuous Functions
on Closed Intervals

Advanced

Calculation

18 min

Q082 · Finding a Complete Range from Bounds and the Intermediate Value Property

Question

Without using derivatives, find the range of $f(x)=x+4/(x+1)$ on $[0,3]$, and explain why every number between the two bounds is attained.

Answer first

The range is $[3,4]$; the minimum 3 occurs at $x=1$, and the maximum 4 occurs at $x=0$ and $x=3$.

Knowledge points

- extrema and intermediate values of continuous functions
- substitution
- algebraic inequalities
- range as a closed interval

Reading the problem and choosing a method

Set $t=x+1$ to transform the interval to $[1,4]$. A square identity gives the lower bound, and the endpoint signs of a factored expression give the upper bound. Continuity then shows that no values between the extrema are missed.

Step-by-step derivation

Step 1. On $[0,3]$, $x+1>0$, so f is continuous. Put $t=x+1$; then $t\in[1,4]$ and $f=t-1+4/t$.

Step 2. For the lower bound, $t+4/t-4=(t-2)^2/t\geq 0$, so $f=t+4/t-1\geq 3$.

Step 3. Equality holds exactly at $t=2$, or $x=1$, so the minimum is 3.

Step 4. For the upper bound, $4-f=5-t-4/t=[-(t-1)(t-4)]/t$.

Step 5. On $t\in[1,4]$, $(t-1)(t-4)\leq 0$ and $t>0$, hence $4-f\geq 0$ and therefore $f\leq 4$.

Step 6. Equality holds at $t=1$ or $t=4$, corresponding to $x=0$ or $x=3$, so the maximum is 4.

Step 7. The function is continuous on the connected interval $[0,3]$ and attains both 3 and 4. By the intermediate value theorem, it attains every value between them.

Step 8. Consequently the complete range is the closed interval $[3,4]$.

Common pitfalls

- finding only the extrema without justifying the full range
- forgetting $t > 0$ when multiplying or dividing inequalities
- missing one endpoint equality case for the upper bound

Verification

Direct values are $f(1)=1+2=3$, $f(0)=4$, and $f(3)=3+1=4$. Continuity rules out gaps inside the range.

Method summary

A continuous-function range problem has two stages: establish attained bounds, then use the intermediate value property to fill the interval between them.

Extension prompt

If the domain were a union of separated sets rather than an interval, continuity alone would not guarantee every value between the global extrema.

§1 · Mappings and Functions

Hard

Error diagnosis

12 min

Textbook-method adaptation

Q083 · Diagnose an Endpoint Error in a Domain

Question

Find the domain of $F(x) = \ln(1 - \sqrt{x-1})$. A student writes: “The radical requires $x \geq 1$. The logarithm requires $1 - \sqrt{x-1} \geq 0$, so $x \leq 2$. Hence the domain is $[1, 2]$.” Identify the error and give the correct solution.

Answer first

The logarithm requires its argument to be strictly positive, not merely nonnegative. The correct domain is $[1, 2)$.

Knowledge points

- domain of a composite function
- strict positivity of a logarithm argument
- nonnegative radicand
- endpoint checking

Reading the problem and choosing a method

The radical condition is correct, but the student weakened the logarithmic requirement from strict positivity to nonnegativity, incorrectly retaining $x=2$ where the logarithm has argument zero.

Step-by-step derivation

Step 1. The inner radical $\sqrt{x-1}$ requires $x-1 \geq 0$, so $x \geq 1$.

Step 2. The outer logarithm requires $1 - \sqrt{x-1} > 0$, not ≥ 0 .

Step 3. Thus $\sqrt{x-1} < 1$. Both sides are nonnegative, so squaring gives $x-1 < 1$, or $x < 2$.

Step 4. Combining $x \geq 1$ and $x < 2$ gives the correct domain $[1, 2)$.

Step 5. At $x=2$, the logarithm argument is $1-1=0$, and $\ln 0$ is undefined; this is the extra endpoint in the student's answer.

Common pitfalls

- confusing the radical condition ≥ 0 with the logarithmic condition > 0
- solving inequalities without substituting candidate endpoints back into the original expression

Verification

At $x=1$, $F(1)=\ln 1=0$ is defined. For $1 < x < 2$ the logarithm argument lies in $(0,1)$. At $x=2$ it becomes zero and must be excluded.

Method summary

Strict inequalities are a major source of domain errors; test every boundary point in the original formula.

Extension prompt

If the outer function were $\sqrt{(1-\sqrt{x-1})}$, a zero outer radicand would be allowed and the domain would indeed be $[1,2]$.

§1 · Mappings and Functions

Hard

Proof

16 min

Q084 · Inverse of a Composition of Bijections

Question

Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be bijections. Prove that $g \circ f:A \rightarrow C$ is a bijection and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Answer first

The composition $g \circ f$ is both injective and surjective, hence bijective. For any $c \in C$, $f^{-1} \circ g^{-1}$ first sends c back to B through g^{-1} and then back to A through f^{-1} . Its compositions with $g \circ f$ in both orders are identity maps, so it equals $(g \circ f)^{-1}$.

Knowledge points

- injectivity and surjectivity
- bijections
- composition
- order of inverse maps

Reading the problem and choosing a method

First prove injectivity and surjectivity separately. The inverse order reverses because one must undo the last-applied map g before undoing the first-applied map f .

Step-by-step derivation

Step 1. Injectivity: if $(g \circ f)(a_1) = (g \circ f)(a_2)$, then $g(f(a_1)) = g(f(a_2))$. Injectivity of g gives $f(a_1) = f(a_2)$, and injectivity of f gives $a_1 = a_2$.

Step 2. Surjectivity: take any $c \in C$. Since g is surjective, some $b \in B$ satisfies $g(b) = c$. Since f is surjective, some $a \in A$ satisfies $f(a) = b$. Thus $(g \circ f)(a) = c$.

Step 3. Therefore $g \circ f$ is both injective and surjective, so it is a bijection from A to C and has an inverse.

Step 4. Let $H = f^{-1} \circ g^{-1}$. For every $c \in C$, $(g \circ f)(H(c)) = g(f(f^{-1}(g^{-1}(c)))) = g(g^{-1}(c)) = c$.

Step 5. For every $a \in A$, $H((g \circ f)(a)) = f^{-1}(g^{-1}(g(f(a)))) = f^{-1}(f(a)) = a$.

Step 6. Since H composes with $g \circ f$ in both orders to give the relevant identity maps, $H = (g \circ f)^{-1}$; hence $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Common pitfalls

- writing the inverse in the unreversed order $g^{-1} \circ f^{-1}$
- claiming invertibility without proving injectivity and surjectivity or checking both compositions

Verification

The set flow is $A \xrightarrow{f} B \xrightarrow{g} C$. Reversing it must give $C \xrightarrow{g^{-1}} B \xrightarrow{f^{-1}} A$, exactly the composition $f^{-1} \circ g^{-1}$.

Method summary

To invert a composition, undo the maps in reverse order: the last operation is undone first.

Extension prompt

Likewise, for three bijections, $(h \circ g \circ f)^{-1} = f^{-1} \circ g^{-1} \circ h^{-1}$.

§2 · Limits of Sequences

Hard

Error diagnosis

20 min

Textbook-method adaptation

Q085 · A Limit Equation Cannot Replace a Convergence Proof

Question

Let $a_1=2$ and $a_{n+1}=\sqrt{3+a_n}$. A student writes: “Assume $a_n \rightarrow L$. Then $L=\sqrt{3+L}$, so $L=(1+\sqrt{13})/2$. Therefore the sequence converges to this value.” Identify the logical error and complete a rigorous proof.

Answer first

The work shows only that, if the sequence converges, its limit must be $(1+\sqrt{13})/2$; it does not prove convergence. Let $L=(1+\sqrt{13})/2$. Induction gives $0 < a_n < L$, and one can prove $a_{n+1} > a_n$. Thus the sequence is increasing and bounded above by L , so it converges. Passing to the limit in the recurrence then gives the limit L .

Knowledge points

- limits of recursive sequences
- monotone bounded principle
- mathematical induction
- a limit equation supplies only candidates

Reading the problem and choosing a method

Passing to a limit assumes that a limit exists. The correct order is to use the positive fixed point as an upper bound, prove monotonicity, invoke monotone bounded convergence, and only then determine the limit.

Step-by-step derivation

Step 1. The gap is treating a necessary condition under an assumed convergence as proof of convergence. Let $L=(1+\sqrt{13})/2$; then $L>2$ and $L^2=L+3$.

Step 2. Upper bound: $a_1=2<L$. If $0 < a_n < L$, then $a_{n+1}=\sqrt{3+a_n} < \sqrt{3+L}=\sqrt{L^2}=L$. Induction gives $0 < a_n < L$.

Step 3. Monotonicity: because $0 < a_n < L$ and $t^2-t-3 < 0$ for $0 < t < L$, we have $a_n^2 < a_n + 3$.

Step 4. Both sides are positive, so taking square roots yields $a_n < \sqrt{a_n + 3} = a_{n+1}$. Hence the sequence is strictly increasing.

Step 5. The sequence is increasing and bounded above by L , so the monotone bounded principle guarantees convergence. Denote its limit by ℓ .

Step 6. Now pass to the limit in the recurrence: $\ell = \sqrt{3 + \ell}$, so $\ell^2 - \ell - 3 = 0$. Since $\ell > 0$, $\ell = (1 + \sqrt{13})/2 = L$.

Common pitfalls

- passing to a limit in a recurrence before proving that the limit exists
- solving the fixed-point equation without rejecting a root incompatible with the signs of the sequence

Verification

The first terms are 2, $\sqrt{5} \approx 2.236$, and $\sqrt{3 + \sqrt{5}} \approx 2.288$; they increase and remain below $L \approx 2.303$. Also $L^2 = L + 3$ verifies the fixed point.

Method summary

For a recursive sequence, prove convergence first and determine the limit second; a fixed-point equation only identifies candidates.

Extension prompt

For recurrences $a_{n+1} = \sqrt{c + a_n}$, first find a positive fixed point and test whether it creates an invariant interval and a useful upper bound.

§3 · Limits of Functions

Hard

Proof

18 min

Textbook-method adaptation

Q086 · Prove Uniqueness of a Function Limit

Question

Let a be an accumulation point of the domain of f . If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} f(x) = B$, prove that $A = B$.

Answer first

Assume $A \neq B$ and choose $\varepsilon = |A - B|/3$. The two limit definitions imply that, in one sufficiently small deleted neighborhood, both $|f(x) - A| < \varepsilon$ and $|f(x) - B| < \varepsilon$ hold. Then $|A - B| \leq |A - f(x)| + |f(x) - B| < 2|A - B|/3$, a contradiction. Hence $A = B$.

Knowledge points

- uniqueness of function limits
- epsilon-delta definition
- taking the minimum of two deltas
- triangle inequality

Reading the problem and choosing a method

If A and B differ, they have a positive separation. Forcing $f(x)$ into neighborhoods of both centers whose radii total less than that separation creates an impossible overlap.

Step-by-step derivation

Step 1. Assume for contradiction that $A \neq B$, and set $d = |A - B| > 0$. Choose $\varepsilon = d/3$.

Step 2. From $\lim_{x \rightarrow a} f(x) = A$, there exists $\delta_1 > 0$ such that $0 < |x - a| < \delta_1$ implies $|f(x) - A| < d/3$.

Step 3. From $\lim_{x \rightarrow a} f(x) = B$, there exists $\delta_2 > 0$ such that $0 < |x - a| < \delta_2$ implies $|f(x) - B| < d/3$.

Step 4. Let $\delta = \min\{\delta_1, \delta_2\}$. Because a is an accumulation point, choose a domain point x with $0 < |x - a| < \delta$; both inequalities then hold simultaneously.

Step 5. By the triangle inequality, $d = |A - B| \leq |A - f(x)| + |f(x) - B| < d/3 + d/3 = 2d/3$.

Step 6. This gives $d < 2d/3$, contradicting $d > 0$. Therefore the assumption is false and $A = B$.

Common pitfalls

- obtaining δ_1 and δ_2 but failing to take their minimum, so the estimates are not guaranteed simultaneously
- forgetting why the accumulation-point condition supplies a domain point in the deleted neighborhood

Verification

Geometrically, the intervals centered at A and B with radius $d/3$ still have a gap of $d/3$, so they cannot contain the same value $f(x)$, matching the algebraic contradiction.

Method summary

For uniqueness, place both limit estimates in one common deleted neighborhood and compare the two candidate limits with the triangle inequality.

Extension prompt

The same proof establishes uniqueness of a sequence limit by combining two thresholds with $\max\{N_1, N_2\}$.

§4 · Infinitesimals
and Infinite Quantities

Hard

Proof

18 min

Textbook-method
adaptation

Q087 · A Bounded Function Times an Infinitesimal

Question

Suppose $\alpha(x)$ is infinitesimal as $x \rightarrow a$ and $g(x)$ is bounded in a deleted neighborhood of a . Using definitions only, prove that $\alpha(x)g(x)$ is also infinitesimal, treating separately the case where the bound may be 0.

Answer first

$\alpha(x)g(x) \rightarrow 0$. If $|g(x)| \leq M$ with $M > 0$, require $|\alpha(x)| < \varepsilon/M$. If $M = 0$, then $g(x) = 0$ throughout that neighborhood and the product is identically 0 there.

Knowledge points

- Epsilon-delta definition of an infinitesimal
- Local boundedness
- Product estimates
- Boundary-case analysis

Reading the problem and choosing a method

The target is to make $|\alpha g|$ smaller than ε . Boundedness supplies a fixed multiplier M , and the case $M = 0$ must be handled before dividing by M .

Step-by-step derivation

Step 1. Since g is locally bounded, there exist $\delta_0 > 0$ and $M \geq 0$ such that $|g(x)| \leq M$ whenever $0 < |x - a| < \delta_0$.

Step 2. If $M = 0$, then $|g(x)| \leq 0$ implies $g(x) = 0$ in that neighborhood, so $\alpha(x)g(x) = 0$ and the result is immediate.

Step 3. Now assume $M > 0$. Given $\varepsilon > 0$, $\alpha(x) \rightarrow 0$ provides $\delta_1 > 0$ such that $|\alpha(x)| < \varepsilon/M$ whenever $0 < |x - a| < \delta_1$.

Step 4. Choose $\delta = \min(\delta_0, \delta_1)$. Both estimates then hold whenever $0 < |x - a| < \delta$.

Step 5. Therefore $|\alpha(x)g(x)| = |\alpha(x)||g(x)| \leq M|\alpha(x)| < M \cdot \varepsilon/M = \varepsilon$.

Step 6. By the definition of an infinitesimal, $\lim_{x \rightarrow a} \alpha(x)g(x) = 0$.

Common pitfalls

- Failing to intersect the two neighborhoods
- Dividing by M when $M=0$
- Citing the conclusion without an epsilon estimate

Verification

For every ε , the proof explicitly selects $\delta = \min(\delta_0, \delta_1)$ and derives $|\alpha g| < \varepsilon$, with exactly the quantifier order required by the definition.

Method summary

For a bounded multiplier, obtain a uniform bound first and scale the epsilon target for the infinitesimal by that bound.

Extension prompt

If $g(x) \rightarrow B$, then g is locally bounded, so this result becomes a key ingredient in proving the product limit law.

§5 · Algebra of Limits

Hard

Error diagnosis

16 min

Textbook-method
adaptation

Q088 · Diagnosing Errors in 0/0 and Cancellation

Question

A student evaluates $\lim_{x \rightarrow 1} (x^2 - 1)/(x - 1)$ as follows: first, substitution gives $0/0=0$; second, canceling $x-1$ gives $x+1$, so the limit is also 2; hence the answer is both 0 and 2. Identify the logical error in each step and give the complete correct solution.

Answer first

The form $0/0$ has no value, so the first step is invalid. Cancellation holds only for $x \neq 1$, but that is exactly enough for a limit. The correct limit is 2.

Knowledge points

- Meaning of an indeterminate form
- Point values versus limits
- Equality in a deleted neighborhood
- Factorization

Reading the problem and choosing a method

The contradiction comes from assigning a numerical value to $0/0$. The cancellation itself is valid, provided one states that it holds for $x \neq 1$, which is sufficient for a limit.

Step-by-step derivation

Step 1. Direct substitution produces only the form $0/0$; this expression is undefined and cannot be set equal to 0.

Step 2. Since $x^2 - 1 = (x - 1)(x + 1)$, for every $x \neq 1$ the quotient equals $x + 1$.

Step 3. The original function may be undefined at $x = 1$, but a limit uses points with $0 < |x - 1| < \delta$, so deleted-neighborhood equality is sufficient.

Step 4. Thus the original limit equals $\lim_{x \rightarrow 1} (x + 1)$.

Step 5. Direct substitution into the polynomial gives $1 + 1 = 2$.

Step 6. The unique correct answer is 2; the value 0 arose from an illegal assignment to $0/0$.

Common pitfalls

- Treating an indeterminate form as a number
- Demanding equality at $x=1$ after cancellation
- Assuming an undefined point prevents a limit

Verification

At $x=0.99$ and $x=1.01$, the original quotient is 1.99 and 2.01, respectively, close to 2 rather than 0.

Method summary

The form $0/0$ is a signal to transform the expression; a limit only requires equality in a deleted neighborhood.

Extension prompt

If two functions agree in some deleted neighborhood of a , then their limits as $x \rightarrow a$ exist together and are equal, regardless of their values at a .

§6 · Existence Criteria and Two Fundamental Limits

Hard

Parameter / synthesis / application

18 min

Q089 · A Variable-Exponent Limit at Infinity

Question

Evaluate $\lim_{x \rightarrow +\infty} (1 + 3/x)^{(2x - \sqrt{x})}$, explicitly identifying the 1^∞ form and the effective standardized exponent.

Answer first

e^6 . With $t=3/x$, the base is $1+t$ and the effective exponent is $t(2x - \sqrt{x}) = 6 - 3/\sqrt{x} \rightarrow 6$.

Knowledge points

- Recognition of 1^∞
- Second fundamental limit
- Variable-exponent standardization
- Composite limits

Reading the problem and choosing a method

The base tends to 1 while the exponent tends to $+\infty$, so the answer is not obtained by direct substitution. The product of the base increment and exponent tends to 6.

Step-by-step derivation

Step 1. As $x \rightarrow +\infty$, $1 + 3/x \rightarrow 1$ and $2x - \sqrt{x} \rightarrow +\infty$, so the form is 1^∞ .

Step 2. Set $t=3/x$. Then $t \rightarrow 0+$ and $(1+t)^{(2x - \sqrt{x})} = [(1+t)^{(1/t)}]^{t(2x - \sqrt{x})}$.

Step 3. The second fundamental limit gives $(1+t)^{(1/t)} \rightarrow e$.

Step 4. The effective exponent is $t(2x - \sqrt{x}) = (3/x)(2x - \sqrt{x}) = 6 - 3/\sqrt{x} \rightarrow 6$.

Step 5. Continuity of the power expression yields the full limit e^6 .

Step 6. This also explains why the base limit 1 alone is insufficient.

Common pitfalls

- Treating 1^∞ as 1
- Ignoring the $-\sqrt{x}$ correction
- Computing $(3/x)^{\sqrt{x}}$ as $3^{\sqrt{x}}$

Verification

At $x=10^4$ the effective exponent is $6-0.03=5.97$, and the expression is close to $e^{5.97}$, moving toward e^6 .

Method summary

A general 1^∞ limit is controlled by the product of the base increment and the exponent.

Extension prompt

Replacing the exponent by $2x-c\sqrt{x}$ for any constant c leaves the limit e^6 because the correction times $3/x$ tends to 0.

§6 · Existence
Criteria and Two
Fundamental Limits

Hard

Proof

25 min

Textbook-method
adaptation

Q090 · A Geometric Proof of the First Fundamental Limit

Question

Assuming radian measure, start from $\sin x < x < \tan x$ for $0 < x < \pi/2$ and prove $\lim_{x \rightarrow 0} \sin x/x = 1$, covering both $x \rightarrow 0^+$ and $x \rightarrow 0^-$.

Answer first

The inequalities give $\cos x < \sin x/x < 1$, so the right-hand limit is 1. Since $\sin(-x)/(-x) = \sin x/x$, the left-hand limit is the same.

Knowledge points

- Unit-circle geometric inequality
- Squeeze theorem
- Even functions
- One-sided limits

Reading the problem and choosing a method

The right-hand limit follows from area comparison and squeezing. The left-hand limit must be handled explicitly by the evenness of $\sin x/x$.

Step-by-step derivation

Step 1. For $0 < x < \pi/2$, comparison of areas in the unit circle gives $\sin x < x < \tan x$.

Step 2. Because $\sin x$ and $\cos x$ are positive, $\sin x < x$ gives $\sin x/x < 1$, while $x < \tan x = \sin x/\cos x$ gives $\cos x < \sin x/x$.

Step 3. As $x \rightarrow 0^+$, $\cos x \rightarrow 1$, so both bounds tend to 1.

Step 4. The squeeze theorem yields $\lim_{x \rightarrow 0^+} \sin x/x = 1$.

Step 5. For $x < 0$, put $u = -x > 0$; then $\sin x/x = \sin(-u)/(-u) = \sin u/u$.

Step 6. Thus the left-hand limit also equals 1, and the two-sided limit is 1.

Common pitfalls

- Not stating that angles are in radians
- Dividing inequalities without checking positivity
- Proving only the right-hand limit
- Calling $\sin x/x$ odd instead of even

Verification

The squeeze can be written $0 < 1 - \sin x/x < 1 - \cos x$, whose right side tends to 0; evenness transfers the result to the negative side.

Method summary

A rigorous proof of the fundamental limit combines geometry, squeezing, and both one-sided limits.

Extension prompt

It immediately implies $\lim_{x \rightarrow 0} \tan x/x = 1$ and $\lim_{x \rightarrow 0} (1 - \cos x)/x^2 = 1/2$.

§7 · Comparison of Infinitesimals

Hard

Proof

18 min

Textbook-method adaptation

Q091 · Equivalence Characterized by Relative Error

Question

Let $\alpha(x)$ and $\beta(x)$ both be infinitesimal as $x \rightarrow a$, with $\beta(x)$ nonzero in a deleted neighborhood. Prove that $\alpha(x) \sim \beta(x)$ if and only if $\alpha(x) - \beta(x) = o(\beta(x))$.

Answer first

Since $[\alpha - \beta]/\beta = \alpha/\beta - 1$, the conditions $\alpha/\beta \rightarrow 1$ and $[\alpha - \beta]/\beta \rightarrow 0$ are exactly equivalent.

Knowledge points

- Definition of equivalent infinitesimals
- Little-o notation
- Biconditional proof
- Relative error

Reading the problem and choosing a method

An exact identity connects the two statements. Both directions must be shown, and eventual nonvanishing of β ensures that every ratio is defined.

Step-by-step derivation

Step 1. Where $\beta(x) \neq 0$, $[\alpha(x) - \beta(x)]/\beta(x) = \alpha(x)/\beta(x) - 1$.

Step 2. If $\alpha \sim \beta$, then $\alpha/\beta \rightarrow 1$ by definition.

Step 3. Hence $[\alpha - \beta]/\beta \rightarrow 1 - 1 = 0$, so $\alpha - \beta = o(\beta)$.

Step 4. Conversely, if $\alpha - \beta = o(\beta)$, then $[\alpha - \beta]/\beta \rightarrow 0$.

Step 5. Using $\alpha/\beta = 1 + [\alpha - \beta]/\beta$ gives $\alpha/\beta \rightarrow 1$.

Step 6. Therefore $\alpha \sim \beta$, proving both directions.

Common pitfalls

- Proving only one direction
- Forgetting eventual nonvanishing of β
- Replacing $o(\beta)$ by merely tending to 0

Verification

Both implications come from the same reversible identity, with the two limits differing exactly by the constant 1.

Method summary

Equivalent infinitesimals have a difference negligible relative to β , not merely an absolute difference tending to 0.

Extension prompt

This criterion explains the common notation $\alpha \sim \beta$ for $\alpha = \beta + o(\beta)$.

§7 · Comparison of Infinitesimals

Hard

Proof

20 min

Textbook-method adaptation

Q092 · Proving Safe Replacement in a Quotient

Question

Suppose $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ as $x \rightarrow a$, with all relevant quotients defined in a deleted neighborhood. Prove $\alpha/\beta \sim \alpha_1/\beta_1$, and explain why this supports equivalent substitution in products and quotients.

Answer first

$$[(\alpha/\beta)/(\alpha_1/\beta_1)] = (\alpha/\alpha_1)(\beta_1/\beta) \rightarrow 1 \cdot 1 = 1.$$

Knowledge points

- Ratio definition of equivalence
- Limits of reciprocals
- Transfer of equivalence through quotients
- Domain conditions

Reading the problem and choosing a method

To compare two quotients for equivalence, take their ratio. It factors into two quantities already known to tend to 1.

Step-by-step derivation

Step 1. From $\alpha \sim \alpha_1$, $\alpha/\alpha_1 \rightarrow 1$; from $\beta \sim \beta_1$, $\beta/\beta_1 \rightarrow 1$.

Step 2. Since the latter limit is $1 \neq 0$, its reciprocal satisfies $\beta_1/\beta \rightarrow 1$.

Step 3. Where all quotients are defined, $(\alpha/\beta)/(\alpha_1/\beta_1) = (\alpha/\alpha_1)(\beta_1/\beta)$.

Step 4. The two factors on the right tend to 1 and 1.

Step 5. Their product tends to 1, so $\alpha/\beta \sim \alpha_1/\beta_1$.

Step 6. Thus factorwise replacement in a product or quotient preserves the final relative ratio.

Common pitfalls

- Not checking eventual nonvanishing of denominators
- Giving β_1/β the wrong limit
- Extending the theorem without conditions to sums or differences

Verification

The proof directly checks that the ratio required by the definition of equivalence tends to 1.

Method summary

Equivalent substitution is safe in products and quotients because relative errors multiply as factors tending to 1.

Extension prompt

The same result for any finite number of product and quotient factors follows by induction.

§7 · Comparison of Infinitesimals

Hard

Proof

20 min

Q093 · A Counterexample to Replacement inside a Sum

Question

Construct and verify a counterexample showing that $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ need not imply $\alpha + \beta \sim \alpha_1 + \beta_1$. Require both sums to be nonzero in a deleted neighborhood.

Answer first

Take $\alpha = x + x^2$, $\alpha_1 = x$, $\beta = -x + x^3$, and $\beta_1 = -x + x^3$. Then $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$, but $(\alpha + \beta)/(\alpha_1 + \beta_1) = (x^2 + x^3)/x^3 = 1/x + 1$, which does not tend to 1.

Knowledge points

- Counterexample construction
- Cancellation of principal terms
- Failure of additive stability
- Ratio verification

Reading the problem and choosing a method

Arrange for first-order terms x and $-x$ to cancel in the sums. The previously negligible higher-order terms then determine the result.

Step-by-step derivation

Step 1. Let $\alpha = x + x^2$ and $\alpha_1 = x$. Then $\alpha/\alpha_1 = 1 + x \rightarrow 1$, so $\alpha \sim \alpha_1$.

Step 2. Let $\beta = \beta_1 = -x + x^3$. They are nonzero in a sufficiently small deleted neighborhood, and $\beta/\beta_1 = 1$.

Step 3. Compute $\alpha + \beta = (x + x^2) + (-x + x^3) = x^2 + x^3 = x^2(1 + x)$.

Step 4. Compute $\alpha_1 + \beta_1 = x + (-x + x^3) = x^3$, nonzero for $x \neq 0$.

Step 5. Their ratio is $[x^2(1 + x)]/x^3 = (1 + x)/x = 1/x + 1$.

Step 6. This ratio does not tend to 1, so the sums are not equivalent.

Common pitfalls

- Using a counterexample whose comparison denominator is identically zero
- Mentioning cancellation without checking the ratio
- Assuming higher-order terms are always disposable

Verification

The two premise ratios tend to 1, while the conclusion ratio is unbounded; the premises hold and the conclusion fails.

Method summary

Addition or subtraction may cancel principal terms, allowing formerly negligible terms to become dominant.

Extension prompt

Replacement in a sum may become safe under an extra condition preventing cancellation and keeping the comparison sum at the original principal order.

§8 · Continuity and Discontinuities

Hard

Parameter / synthesis / application

28 min

Q094 · Can a Power Factor Suppress Oscillation?

Question

For a real parameter p , define $f_p(0)=0$ and $f_p(x)=|x|^p \sin(1/x)$ for $x \neq 0$. Discuss continuity at $x=0$. When it is discontinuous, classify the discontinuity and decide whether the function is bounded in some neighborhood of 0.

Answer first

For $p > 0$, it is continuous at 0 and locally bounded; for $p = 0$, it has a bounded oscillatory discontinuity of the second kind; for $p < 0$, it has an unbounded oscillatory discontinuity of the second kind.

Knowledge points

- squeeze theorem
- oscillatory functions
- parameter case analysis
- sequential test for limits
- local boundedness

Reading the problem and choosing a method

The decisive quantity is the amplitude $|x|^p$. For $p > 0$ it tends to zero and suppresses the sine oscillation; for $p = 0$ it stays fixed; for $p < 0$ it grows without bound. In the last two cases, explicit sequences rigorously disprove the existence of a limit.

Step-by-step derivation

Step 1. If $p > 0$, then $|f_p(x)| = |x|^p |\sin(1/x)| \leq |x|^p \rightarrow 0$. The squeeze theorem gives $\lim_{x \rightarrow 0} f_p(x) = 0 = f_p(0)$, so the function is continuous.

Step 2. Still for $p > 0$, if $|x| < 1$, then $|f_p(x)| \leq |x|^p < 1$, so the function is locally bounded.

Step 3. If $p = 0$, then $f_0(x) = \sin(1/x)$. For $x_n = 1/(\pi/2 + 2\pi n)$, $f_0(x_n) = 1$; for $y_n = 1/(3\pi/2 + 2\pi n)$, $f_0(y_n) = -1$.

Step 4. Since $x_n, y_n \rightarrow 0$ but the function values approach different numbers, the limit fails to exist. This is an oscillatory discontinuity of the second kind, while $|f_0(x)| \leq 1$ proves local boundedness.

Step 5. If $p < 0$, use $x_n = 1/(\pi/2 + 2\pi n)$ again. Then $f_p(x_n) = |x_n|^p \rightarrow +\infty$.

Step 6. Now take $z_n = 1/(\pi n)$. Since $\sin(1/z_n) = 0$, we have $f_p(z_n) = 0$. Two sequences approaching 0 produce infinite and zero behavior, so no single limit exists.

Step 7. The divergence $f_p(x_n) \rightarrow +\infty$ also proves unboundedness in every neighborhood of 0. Thus $p < 0$ gives an unbounded oscillatory discontinuity of the second kind.

Step 8. Combining the three parameter ranges yields the complete classification.

Common pitfalls

- assuming $|\sin(1/x)| \leq 1$ forces the product to zero for every p
- writing the limit as infinity for $p < 0$ while ignoring zeros and sign changes of the sine factor
- misclassifying bounded oscillation as a first-kind discontinuity

Verification

The boundary case $p = 0$ is treated separately. A uniform bound proves the $p > 0$ result, while explicit sequences for $p = 0$ and $p < 0$ both disprove the limit and establish the stated boundedness behavior.

Method summary

A vanishing amplitude times a bounded oscillation is handled by squeezing; if the amplitude does not vanish, reveal the oscillation with carefully chosen sequences.

Extension prompt

If $\sin(1/x)$ is replaced by any bounded function $h(1/x)$, the conclusion for $p > 0$ remains true; for $p \leq 0$, the behavior depends on the actual oscillation of h .

§10 · Continuous Functions on Closed Intervals

Hard

Proof

22 min

Textbook-method adaptation

Q095 · Equal Values Half an Interval Apart

Question

Suppose f is continuous on $[0, 1]$ and $f(0)=f(1)$. Prove that there is at least one $\xi \in [0, 1/2]$ such that $f(\xi)=f(\xi+1/2)$.

Answer first

Define $g(x)=f(x)-f(x+1/2)$ for $0 \leq x \leq 1/2$. Then $g(1/2)=-g(0)$. If an endpoint value is zero, the result is immediate; otherwise the endpoint values have opposite signs and the zero theorem supplies the required ξ .

Knowledge points

- construction of an auxiliary function
- difference of continuous functions
- zero theorem
- opposite endpoint values

Reading the problem and choosing a method

The desired equation compares values of the same function at points separated by $1/2$. Moving the two sides into a difference converts the problem into finding a zero. The condition $f(0)=f(1)$ makes the endpoint values of this difference exact negatives.

Step-by-step derivation

Step 1. Define $g(x)=f(x)-f(x+1/2)$ on $[0, 1/2]$.

Step 2. For $x \in [0, 1/2]$, the shifted point $x+1/2$ belongs to $[1/2, 1]$. Since f is continuous, both $f(x)$ and $f(x+1/2)$ are continuous, so g is continuous.

Step 3. At the left endpoint, $g(0)=f(0)-f(1/2)$.

Step 4. At the right endpoint, $g(1/2)=f(1/2)-f(1)=f(1/2)-f(0)=-g(0)$, using $f(0)=f(1)$.

Step 5. If $g(0)=0$, choose $\xi=0$ and obtain $f(0)=f(1/2)$; in this case $g(1/2)$ is also zero.

Step 6. If $g(0) \neq 0$, then $g(0)$ and $g(1/2)=-g(0)$ have opposite signs.

Step 7. The zero theorem gives some $\xi \in (0, 1/2)$ with $g(\xi)=0$.

Step 8. In either case, some $\xi \in [0, 1/2]$ satisfies $f(\xi)-f(\xi+1/2)=0$, which is the desired equality.

Common pitfalls

- applying the zero theorem directly to f without constructing a difference
- failing to check that $x+1/2$ stays in the domain of f
- handling only strict opposite signs and omitting the endpoint-zero case

Verification

Continuity of the auxiliary function, its endpoint relation, and every hypothesis of the zero theorem have been checked; the equation $g(\xi)=0$ is exactly equivalent to the target statement.

Method summary

For equal-value existence problems, form the difference of the two sides and seek a sign relation at the endpoints.

Extension prompt

For an integer $n \geq 2$, one can similarly study differences $f(x)-f(x+1/n)$ and use cyclic relations among subdivision values.

§10 · Continuous Functions
on Closed Intervals

Hard

Parameter / synthesis /
application

30 min

Q096 · Optional: Comparing Uniform Continuity on an Unbounded Interval

Question

On $[0, +\infty)$, determine whether $f(x) = \sqrt{x}$ and $g(x) = x^2$ are uniformly continuous. Prove both conclusions directly from the definition. For the function that is not uniformly continuous, give a specific ε_0 and a contradiction construction.

Answer first

$f(x) = \sqrt{x}$ is uniformly continuous on $[0, +\infty)$ with $\delta = \varepsilon^2$; $g(x) = x^2$ is not uniformly continuous there, as shown by $\varepsilon_0 = 1$ and $x = n, y = n + 1/n$.

Knowledge points

- definition of uniform continuity
- estimate for differences of square roots
- negation of uniform continuity
- unbounded interval

Reading the problem and choosing a method

Uniform continuity requires one δ to work for every pair of points in the interval. The global estimate $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$ provides uniform control. The square function becomes increasingly steep far out, allowing pairs whose input distance tends to zero while the output difference stays above a fixed number.

Step-by-step derivation

Step 1. First prove the square-root estimate. For $x, y \geq 0$, assume without loss of generality that $x \geq y$. Then

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Step 2. Hence for all $x, y \geq 0$, $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$.

Step 3. Given $\varepsilon > 0$, choose $\delta = \varepsilon^2$. If $|x - y| < \delta$, then $|f(x) - f(y)| \leq \sqrt{|x - y|} < \sqrt{\delta} = \varepsilon$.

Step 4. This δ depends only on ε , not on the particular points, so $\sqrt{x}$ is uniformly continuous on $[0, +\infty)$.

Step 5. Now consider $g(x)=x^2$. To negate uniform continuity, fix $\varepsilon_0=1$.

Step 6. Given any $\delta>0$, choose a positive integer $n>1/\delta$, and set $x=n$ and $y=n+1/n$. Then $|x-y|=1/n<\delta$.

Step 7. However, $|g(y)-g(x)|=(n+1/n)^2-n^2=2+1/n^2>1=\varepsilon_0$.

Step 8. Thus every proposed δ is defeated by a pair closer than δ , so x^2 is not uniformly continuous on $[0,+\infty)$.

Common pitfalls

- using a point-dependent delta from ordinary continuity as though it were uniform
- concluding from an unbounded derivative instead of following the requested definition
- allowing epsilon to vary with n when negating uniform continuity

Verification

For the square root, $\delta=\varepsilon^2$ works uniformly over the whole interval. For the square, the same fixed $\varepsilon_0=1$ defeats every δ through an explicit pair, exactly matching the quantified negation.

Method summary

Uniform continuity is global uniform control; on an unbounded interval, ordinary continuity does not automatically imply it.

Extension prompt

A similar estimate proves that x^α is uniformly continuous on $[0,+\infty)$ for $0<\alpha\leq 1$, while the square-function construction adapts when $\alpha>1$.

Challenge

Attempt independently before consulting the solutions.

§2 · Limits of Sequences

Challenge

Parameter / synthesis /
application

28 min

Q097 · Characterize Parameter Convergence Using Even and Odd Subsequences

Question

For real p and q , define $a_{2k} = p + 1/k$ and $a_{2k-1} = q - 1/k$ for $k=1, 2, \dots$. Find a necessary and sufficient condition for $\{a_k\}$ to converge. In the convergent case, state the limit and prove it using the epsilon- N definition.

Answer first

$\{a_k\}$ converges if and only if $p=q$; in that case its limit is the common value $p=q$.

Knowledge points

- even and odd subsequences
- subsequences inherit the original limit
- proof of an if-and-only-if statement
- uniform control of both index classes

Reading the problem and choosing a method

Necessity follows because the two natural subsequences of a convergent sequence must have the same limit. For sufficiency, one should provide a single N that controls both even and odd terms.

Step-by-step derivation

Step 1. Necessity: if $\{a_k\}$ converges to A , then both its even and odd subsequences must also converge to A .

Step 2. Since $a_{2k} = p + 1/k$, $\lim_{k \rightarrow \infty} a_{2k} = p$. Since $a_{2k-1} = q - 1/k$, $\lim_{k \rightarrow \infty} a_{2k-1} = q$. Thus $p=q=A$ is necessary.

Step 3. Sufficiency: suppose $p=q=r$. Given $\epsilon > 0$, choose a positive integer $K > 1/\epsilon$ and set $N=2K$.

Step 4. If $n > N$ is even, write $n=2k$. Then $k > K$, so $|a_n - r| = 1/k < \epsilon$.

Step 5. If $n > N$ is odd, write $n=2k-1$. From $2k-1 > 2K$ we get $k > K+1/2$, hence $k > K$ and $|a_n - r| = 1/k < \epsilon$.

Step 6. Thus $|a_n - r| < \varepsilon$ for every $n > N$, regardless of parity. The epsilon- N definition gives $a_n \rightarrow r$. Therefore convergence occurs exactly when $p = q$, with that common value as the limit.

Common pitfalls

- computing the two subsequence limits without stating why convergence of the original forces them to agree
- choosing separate thresholds for even and odd terms without combining them into one N

Verification

If $p \neq q$, choose $\varepsilon = |p - q|/4$. Sufficiently large even terms lie near p and odd terms near q ; these neighborhoods are disjoint, so the full sequence cannot approach one limit.

Method summary

For a parity-defined sequence, use subsequences to obtain necessary parameter conditions, then prove sufficiency with one tail estimate covering both parities.

Extension prompt

If $1/k$ is replaced by any positive sequence $b_n \rightarrow 0$, the same condition $p = q$ remains valid; use $b_n \rightarrow 0$ in place of the explicit reciprocal estimate.

§6 · Existence
Criteria and Two
Fundamental Limits

Challenge

Parameter / synthesis
/ application

32 min

Textbook-method
adaptation

Q098 · A Complete Convergence Chain for a Radical Recurrence

Question

Let $0 < a_1 < 2$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove: (1) $0 < a_n < 2$; (2) $\{a_n\}$ is increasing; (3) it converges; and (4) find its limit. Do not guess a closed form or merely solve a limit equation.

Answer first

For every n , $0 < a_n < 2$ and $a_{n+1} > a_n$. The sequence is increasing and bounded above by 2, so it converges to 2.

Knowledge points

- Invariant interval for a recurrence
- Mathematical induction
- Comparison by squaring
- Monotone convergence theorem
- Filtering roots of a limit equation

Reading the problem and choosing a method

First prove that $(0, 2)$ is invariant under the recurrence. Inside this interval, $a_{n+1} > a_n$ is equivalent to $2 + a_n > a_n^2$, whose sign factors cleanly.

Step-by-step derivation

Step 1. The base case has $0 < a_1 < 2$. If $0 < a_n < 2$, then $2 < 2 + a_n < 4$, hence $\sqrt{2} < a_{n+1} < 2$; induction gives $0 < a_n < 2$ for all n .

Step 2. Both a_n and a_{n+1} are positive, so their order may be compared by squaring.

Step 3. Compute $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(a_{n+1}) > 0$.

Step 4. Thus $a_{n+1} > a_n$. The sequence is increasing and, from the first step, bounded above by 2.

Step 5. The monotone convergence theorem gives convergence. Let the limit be L , with $0 < L \leq 2$.

Step 6. Passing to the limit gives $L = \sqrt{2+L}$, hence $L^2 - L - 2 = 0$ and $L = 2$ or $L = -1$.

Step 7. Since $L > 0$, reject -1 and obtain $L = 2$.

Common pitfalls

- Solving $L = \sqrt{2+L}$ without first proving convergence
- Squaring without confirming nonnegativity
- Keeping both algebraic roots
- Proving only the upper bound in the induction

Verification

For $a_1 = 1$, the terms $1, \sqrt{3}, \sqrt{2+\sqrt{3}}$ remain in $(0, 2)$ and increase; the limit 2 satisfies the recurrence equation.

Method summary

For nonlinear recurrences: invariant interval, monotonicity, boundedness, existence, limit equation, and root filtering.

Extension prompt

If $a_1 > 2$, one can instead prove the sequence decreases and is bounded below by 2, again converging to 2.

§7 · Comparison of
Infinitesimals

Challenge

Parameter / synthesis
/ application

35 min

Textbook-method
adaptation

Q099 · Safe and Unsafe Zones for Equivalent Substitution

Question

Complete the following without derivatives, L'Hôpital's rule, or Taylor expansions. (1) Evaluate $A = \lim_{x \rightarrow 0} \sin(2x)/(e^x - 1)$. (2) A student uses $\sqrt{1+x} - 1 \sim x/2$ to replace $\sqrt{1+x} - 1$ inside a difference and claims $B = \lim_{x \rightarrow 0} [\sqrt{1+x} - 1 - x/2]/x^2 = 0$. Judge the method and conclusion, then find B by exact algebra. (3) State one rule describing when direct equivalent-infinitesimal substitution is valid.

Answer first

$A=2$. The student's method is invalid and the conclusion is false; $B=-1/8$. Direct substitution is safe for multiplicative and divisive factors, but not mechanically inside sums or differences where principal terms may cancel.

Knowledge points

- Equivalent substitution in products and quotients
- Cancellation in differences
- Exact radical rationalization
- Method-validity audit

Reading the problem and choosing a method

Part (1) has a safe quotient structure. Part (2) magnifies the second-order error discarded by a first-order equivalence, so the exact remainder must be retained.

Step-by-step derivation

Step 1. For A, $\sin(2x) \sim 2x$ and $e^x - 1 \sim x$. These are quotient factors, so $A = \lim_{x \rightarrow 0} 2x/x = 2$.

Step 2. For B, replacing one term inside a difference makes equal first-order terms cancel; the equivalence contains no information about the remaining second-order term, so the method is invalid.

Step 3. Set $r = \sqrt{1+x}$. Then $x = r^2 - 1 = (r-1)(r+1)$, and $r \rightarrow 1$.

Step 4. The numerator is $r - 1 - (r^2 - 1)/2 = (r-1)[1 - (r+1)/2] = -(r-1)^2/2$.

Step 5. Since $x^2 = (r-1)^2(r+1)^2$, cancellation for $x \neq 0$ gives $-1/[2(r+1)^2]$.

Step 6. Letting $r \rightarrow 1$ yields $B = -1/[2 \cdot (2)^2] = -1/8$, so the numerical claim is also false.

Step 7. Rule: replace equivalent infinitesimals factorwise in products and quotients; in sums or differences with possible cancellation, use exact algebra or higher-order remainder information.

Common pitfalls

- Treating equivalence as identity
- Ignoring cancellation of principal terms
- Using Taylor expansions outside the allowed chapter methods
- Losing a square during rationalization

Verification

A factors as $[\sin 2x/(2x)] \cdot [x/(e^x - 1)] \cdot 2 \rightarrow 2$. For B, the exact expression $-1/[2(\sqrt{1+x}+1)^2]$ tends to $-1/8$.

Method summary

Equivalence preserves first-order relative information; once subtraction cancels that first-order term, exact algebra or higher-order information is required.

Extension prompt

The same danger appears in $\infty - \infty$ forms: subtracting separate leading equivalents may erase the information that determines the limit.

§10 · Continuous Functions
on Closed Intervals

Challenge

Parameter / synthesis /
application

40 min

Q100 · Optional: The Uniform-Continuity Boundary for Oscillatory Functions

Question

On $(0, 1]$, define $F(x) = x \sin(1/x)$ and $G(x) = \sin(1/x)$. Determine whether each is uniformly continuous. For F , use a continuous extension and the uniform-continuity theorem on a closed interval. For G , construct two sequences whose distance tends to 0 while their function values differ by exactly 2.

Answer first

F is uniformly continuous on $(0, 1]$ because defining $F(0) = 0$ gives a continuous function on $[0, 1]$. G is not uniformly continuous; take $x_n = 1/(\pi/2 + 2\pi n)$ and $y_n = 1/(3\pi/2 + 2\pi n)$.

Knowledge points

- continuous extension
- uniform continuity of a continuous function on a closed interval
- squeeze theorem
- counterexample sequences for uniform continuity
- rapid oscillation

Reading the problem and choosing a method

Both functions oscillate increasingly rapidly, but the amplitude x in F tends to zero and permits a continuous extension at 0. The closed-interval theorem then gives uniform continuity. The amplitude of G does not decay, so increasingly close peaks and troughs remain two units apart in value.

Step-by-step derivation

Step 1. For F , $|x \sin(1/x)| \leq x$. As $x \rightarrow 0^+$, the right-hand side tends to 0, so the squeeze theorem gives $\lim_{x \rightarrow 0^+} F(x) = 0$.

Step 2. Define $F(0) = 0$ and $F(x) = F(x)$ on $(0, 1]$. The original expression is continuous at positive points, and the previous limit proves right-continuity at 0.

Step 3. Thus F is continuous on the closed interval $[0, 1]$. By the theorem that a continuous function on a closed interval is uniformly continuous, F is uniformly continuous on $[0, 1]$.

Step 4. A restriction of a uniformly continuous function to any subset is uniformly continuous, so F is uniformly continuous on $(0, 1]$.

Step 5. For G , set $x_n = 1/(\pi/2 + 2\pi n)$ and $y_n = 1/(3\pi/2 + 2\pi n)$. For all sufficiently large n , both lie in $(0, 1]$, and both tend to 0.

Step 6. We have $G(x_n) = 1$ and $G(y_n) = -1$, so $|G(x_n) - G(y_n)| = 2$ for every such n .

Step 7. Meanwhile, $|x_n - y_n| = \pi/[(\pi/2 + 2\pi n)(3\pi/2 + 2\pi n)] \rightarrow 0$.

Step 8. If G were uniformly continuous, choosing $\varepsilon = 1$ would give a $\delta > 0$ forcing every pair less than δ apart to have value difference below 1. Large n violate this with distance below δ but value difference 2.

Step 9. Therefore G is not uniformly continuous. The distinction is whether the oscillation amplitude decays enough to allow a continuous endpoint extension.

Common pitfalls

- declaring both functions non-uniformly continuous merely because they oscillate rapidly
- proving only pointwise continuity of F without obtaining a uniform result
- choosing two sequences without checking both mutual distance and a fixed output separation

Verification

The extension of F satisfies every hypothesis of the closed-interval theorem. For G , the point-distance formula tends to 0 while the value difference stays exactly 2, directly contradicting uniform continuity with fixed $\varepsilon = 1$.

Method summary

Infinite oscillation alone need not destroy uniform continuity; decay of amplitude and continuous extendability at the missing endpoint are decisive.

Extension prompt

For $x^p \sin(1/x)$ on $(0, 1]$, every $p > 0$ allows a continuous extension to $[0, 1]$ and hence uniform continuity, while $p = 0$ gives the present function G .

Training value, limitations, and second-pass route

Strengths

- All ten sections are covered and indexed by knowledge point.
- Conceptual, computational, proof, parameter, counterexample, and error-diagnosis tasks expose more than routine algebra.
- Even the hard questions stay within Chapter 1 tools and train the real boundaries of squeeze arguments, equivalent infinitesimals, and continuity theorems.
- Chinese and English identifiers and mathematics match one-to-one, supporting calculus vocabulary development.

Limitations

- One hundred questions cannot exhaust every algebraic transformation; weak categories may need extra drills.
- Textbook material is represented by independently rewritten method adaptations, not verbatim textbook questions.
- Difficulty is partly subjective and depends on prior algebra preparation.
- Uniform continuity is starred enrichment; skip those optional items if your course omits it.
- To respect the Chapter 1 boundary, later shortcuts such as L'Hopital's rule and Taylor expansions are not shown.

Recommended second pass

1. First pass: work under time limits, record confidence, and do not open the solutions.
2. Correction: classify each error as conceptual, algebraic, method-selection, or rigor/communication.
3. After 48 hours: redo wrong items and their neighbors without prompts.
4. After one week: sample by knowledge tag, emphasizing limit definitions, cancellation in equivalent infinitesimals, and theorem hypotheses.